

#1 · Common Occupational Skills

What does due diligence mean for a crane operator?

Answer

It means taking every reasonable step to prevent harm, such as inspecting equipment and confirming ground conditions. It is more than following a checklist. Documented due diligence can protect the operator if an incident happens anyway.

#2 · Common Occupational Skills

What is the right to refuse unsafe work?

Answer

Every worker, including crane operators, can stop a task they believe is unsafe without being punished for it. The correct steps are to stop, report the specific reason, and wait for investigation before resuming.

#3 · Common Occupational Skills

What is the internal responsibility system?

Answer

It is the idea that everyone on site shares specific safety responsibilities, from labourers to supervisors. Responsibility is not concentrated in one role or diluted across the group.

#4 · Common Occupational Skills

What is a lockout and tagout procedure for?

Answer

It isolates a crane's energy sources before maintenance so the machine cannot move unexpectedly. A tag identifies who applied the lock, and only that person should remove it.

#5 · Common Occupational Skills

Why is fall protection critical for crane operators?

Answer

Boom climbing, mast erection, and elevated cab access all involve fall risk. A full body harness with a shock-absorbing lanyard connected to an approved anchor point is the standard setup.

#6 · Common Occupational Skills

What does 100% tie-off mean?

Answer

A worker stays connected to an approved anchor point at all times while at height, using two lanyards so one stays connected while the other is repositioned.

#7 · Common Occupational Skills

What is the hierarchy of controls?

Answer

It ranks hazard controls from elimination first, then engineering controls, then administrative controls, with PPE last. PPE is the final layer of protection, not the first choice.

#8 · Common Occupational Skills

What does the WHMIS flame pictogram warn about?

Answer

A black flame symbol inside a red diamond border warns of flammable materials, such as fuels and solvents commonly found on a crane site.

#9 · Common Occupational Skills

What is a safety data sheet (SDS)?

Answer

A standardized document required under WHMIS 2015 that explains a controlled product's hazards, handling, storage, and first aid measures.

#10 · Common Occupational Skills

What is the difference between a supplier label and a workplace label?

Answer

A supplier label comes from the manufacturer with the pictogram and hazard statements. A workplace label is created when a product is transferred to a new container on site and must still show hazard information.

#11 · Common Occupational Skills

Which fire extinguisher class fits a fuel or hydraulic fluid fire?

Answer

Class B extinguishers are rated for flammable liquid fires, including fuel and hydraulic oil common around crane engines.

#12 · Common Occupational Skills

Which fire extinguisher class fits an energized electrical fire?

Answer

Class C extinguishers are rated for fires involving energized electrical equipment, such as wiring or electrical components.

#13 · Common Occupational Skills

What does the PASS method stand for?

Answer

Pull the pin, aim at the base of the fire, squeeze the handle, and sweep side to side. This is the standard sequence for using a fire extinguisher.

#14 · Common Occupational Skills

When should a fire not be fought with an extinguisher?

Answer

When the fire is too large or the extinguisher is not the correct class for the fuel involved. Evacuate and call for help instead of continuing to fight it.

#15 · Common Occupational Skills

What should happen to a cracked or damaged hand tool?

Answer

It should be removed from service immediately, since a damaged tool can fail under load and cause injury.

#16 · Common Occupational Skills

Why does housekeeping matter on a crane site?

Answer

Tools or debris left near the swing radius or walkways create tripping and impact hazards, especially near moving equipment.

#17 · Common Occupational Skills

What does an equipment out-of-service tag mean?

Answer

The tagged tool or hardware must never be used, even for a small job, until it is properly cleared or replaced by qualified personnel.

#18 · Common Occupational Skills

What does a daily crane log record?

Answer

Pre-operational inspection results, hours of operation, and any issues noted during the shift, creating a documented history that supports due diligence.

#19 · Common Occupational Skills

What is a near miss and why report it?

Answer

A near miss is an event that could have caused harm but did not. Reporting it helps identify hazards before they cause real injury.

#20 · Common Occupational Skills

What information does a written lift plan document?

Answer

Load weight, radius, crane configuration, and site conditions for a planned lift. The operator should confirm it matches actual conditions before lifting.

#21 · Common Occupational Skills

What is a mentor's responsibility when assigning tasks to an apprentice?

Answer

Match tasks to the apprentice's current skill level and supervise closely as new or unfamiliar tasks are introduced.

#22 · Common Occupational Skills

How should a mentor handle an apprentice's unsafe habit?

Answer

Correct it immediately, calmly, and clearly, rather than waiting until later, since delayed correction lets the habit become permanent.

#23 · Common Occupational Skills

What is the hand signal for hoist?

Answer

Raise a forearm with the index finger pointing up and move it in a small circle, signaling the load should be raised.

#24 · Common Occupational Skills

What is the hand signal for emergency stop and who can give it?

Answer

Both arms extended and moved rapidly. Any worker can give this signal at any time, and the operator must obey it immediately.

#25 · Common Occupational Skills

What is the hand signal for dog everything (hold)?

Answer

Both hands clasped together in front of the body, meaning stop and hold the current position.

#26 · Common Occupational Skills

How many signal persons should direct an operator at once?

Answer

Only one, except for the emergency stop signal, which any worker can give at any time regardless of who is the designated signal person.

#27 · Common Occupational Skills

What should an operator do if radio and hand signals conflict?

Answer

Stop the lift and confirm communication before continuing, rather than guessing which signal to trust.

#28 · Common Occupational Skills

What is good radio protocol during a lift with limited visibility?

Answer

Use the signal person's name or role, and confirm instructions by repeating them back before acting on them.

#29 · Common Occupational Skills

What should be placed under a crane during fueling to protect the environment?

Answer

A drip pan or absorbent material, to catch spilled fuel before it can reach soil or a nearby waterway.

#30 · Common Occupational Skills

What does a site orientation cover and when is it required?

Answer

It covers hazards specific to that site, including traffic patterns, exclusion zones, and utilities. It is required at every new site, even for experienced operators.

#31 · Hoisting Calculations

What is the formula for tension in each leg of a symmetric two-leg bridle sling?

Answer

Tension per leg = load / (2 x sin of the angle from horizontal). As the angle decreases toward horizontal, tension in each leg rises sharply.

#32 · Hoisting Calculations

What is the sling leg tension multiplier for a 30 degree sling angle from horizontal?

Answer

The multiplier is 2.000, meaning each leg carries twice the tension it would at a 90 degree (vertical) angle. This is the steepest commonly tabulated shallow angle used in rigging.

#33 · Hoisting Calculations

What is the sling leg tension multiplier for a 45 degree sling angle from horizontal?

Answer

The multiplier is approximately 1.414 (the square root of 2). Tension per leg = (load/legs) x 1.414 at this angle.

#34 · Hoisting Calculations

What is the sling leg tension multiplier for a 60 degree sling angle from horizontal?

Answer

The multiplier is approximately 1.155. This is a commonly recommended angle because it keeps leg tension only slightly above the load's share.

#35 · Hoisting Calculations

Why do riggers avoid sling angles below 30 degrees from horizontal?

Answer

Below 30 degrees, leg tension increases very rapidly for small angle decreases, risking sling failure. Most rigging standards treat angles under 30 degrees as unacceptable without engineering review.

#36 · Hoisting Calculations

What is the formula for operating radius using boom length and boom angle?

Answer

Radius = boom length x cosine of the boom angle (measured from horizontal). This gives the horizontal distance from the center of rotation to the boom tip.

#37 · Hoisting Calculations

What is the formula for boom tip height using boom length and boom angle?

Answer

Height = boom length x sine of the boom angle, plus the height of the boom foot pin above ground. This tells the operator how high the boom tip sits for clearance planning.

#38 · Hoisting Calculations

As boom angle increases toward vertical (90 degrees), what happens to operating radius?

Answer

Operating radius decreases, since radius = boom length x cos(angle) and cosine approaches zero near 90 degrees. A near-vertical boom gives the shortest possible radius for that boom length.

#39 · Hoisting Calculations

As boom angle decreases toward horizontal (0 degrees), what happens to operating radius?

Answer

Operating radius increases toward its maximum, approaching the full boom length. A low boom angle gives the longest reach but the lowest rated capacity.

#40 · Hoisting Calculations

What two values does an operator use to look up rated capacity on a mobile crane load chart?

Answer

Boom length (column) and operating radius (row). The cell where they intersect is the maximum rated capacity for that configuration.

#41 · Hoisting Calculations

What is the general trend between operating radius and rated capacity on any load chart?

Answer

Rated capacity decreases as radius increases, for a fixed boom length. This is true whether the limiting factor is tipping or structural strength.

#42 · Hoisting Calculations

What does the term structural capacity mean on a crane load chart?

Answer

The maximum load the boom and machinery can physically withstand before deforming or failing, independent of tipping stability. It typically governs at shorter operating radii.

#43 · Hoisting Calculations

What does the term tipping capacity mean on a crane load chart?

Answer

The maximum load the crane can lift before it becomes unstable and begins to tip over its fulcrum point. It typically governs at longer operating radii.

#44 · Hoisting Calculations

At short operating radii, which limit usually governs rated capacity: tipping or structural?

Answer

Structural capacity usually governs at short radii, since the crane is stable but the boom and machinery are highly stressed. The chart often marks these cells differently (e.g. bold or asterisked).

#45 · Hoisting Calculations

At long operating radii, which limit usually governs rated capacity: tipping or structural?

Answer

Tipping capacity usually governs at long radii, since the load's moment arm makes the crane more prone to tipping before the structure itself is overstressed.

#46 · Hoisting Calculations

What is ground bearing pressure (GBP) and how is it calculated?

Answer

GBP is the pressure a loaded outrigger exerts on the ground, calculated as reaction load divided by the pad's contact area. It is usually expressed in pounds per square foot (psf) or psi.

#47 · Hoisting Calculations

Why must ground bearing pressure be checked before every outrigger set-up?

Answer

If GBP exceeds what the soil can support, the outrigger pad can sink or fail, tipping the crane. Mats or cribbing are added to spread the load over a larger area and reduce pressure.

#48 · Hoisting Calculations

How does increasing outrigger pad size affect ground bearing pressure for the same load?

Answer

Increasing pad area decreases GBP, since $GBP = \text{load} / \text{area}$. Doubling the contact area cuts the pressure roughly in half for the same reaction load.

#49 · Hoisting Calculations

What is the formula for line pull in a hoist line reeved with multiple parts, ignoring friction?

Answer

Line pull per part = total suspended load / number of parts of line. More parts of line means lower tension per part but slower hook travel for the same drum speed.

#50 · Hoisting Calculations

Why does reeving a hoist line with more parts of line reduce wire rope tension?

Answer

Adding parts of line spreads the same total load across more strands, giving mechanical advantage. This lets a crane lift heavier loads without exceeding the rope's rated breaking strength.

#51 · Hoisting Calculations

What is the tradeoff of using more parts of line to gain mechanical advantage?

Answer

More parts of line reduces line pull but also reduces hook block travel speed for a given drum rotation speed. Heavier lifts require slower, more deliberate hoisting.

#52 · Hoisting Calculations

How does sheave friction affect actual line pull compared to the frictionless calculation?

Answer

Friction increases the actual line pull needed above the simple load-divided-by-parts estimate, since some of the drum's pulling force is lost overcoming sheave resistance. Reeving efficiency tables account for this loss.

#53 · Hoisting Calculations

What three items are commonly deducted from a load chart's gross capacity to find net lifting capacity?

Answer

Hook block (or headache ball) weight, rigging weight (slings, shackles, spreader bars), and jib weight if the jib is erected. All three reduce how much actual payload the crane can lift.

#54 · Hoisting Calculations

Why is jib weight deducted from capacity even if the jib itself is not making the pick?

Answer

An erected jib still adds weight and moment to the boom system that must be supported, whether or not it is used for the current pick. Charts for jib-erected configurations already build this deduction into their capacity figures.

#55 · Hoisting Calculations

What is net lifting capacity and how is it found?

Answer

Net capacity is the actual usable payload capacity, found by subtracting hook block, rigging, and jib (if erected) weight from the chart's gross rated capacity. The load itself must not exceed this net figure.

#56 · Hoisting Calculations

What is meant by operating quadrant on a mobile crane?

Answer

The direction relative to the carrier that a lift is made in, typically over-front, over-side, or over-rear. Load charts often list different rated capacities for each quadrant at the same radius.

#57 · Hoisting Calculations

Why is over-side rated capacity usually lower than over-front capacity at the same radius?

Answer

The crane's stability fulcrum is generally narrower when lifting over the side compared to over the front on outriggers. This reduced stability margin lowers the tipping capacity for over-side lifts.

#58 · Hoisting Calculations

Why must an operator recheck the load chart when swinging a suspended load from one quadrant to another?

Answer

Quadrant is a separate chart variable from radius and boom angle, and capacity can drop significantly even if radius and angle stay the same. Failing to recheck can result in exceeding rated capacity mid-swing.

#59 · Hoisting Calculations

How does wind speed generally affect a mobile crane's rated lifting capacity?

Answer

Increasing wind speed reduces rated capacity, and most manufacturers require a percentage capacity reduction above a stated threshold speed. Operations must stop entirely above the manufacturer's maximum rated wind speed.

#60 · Hoisting Calculations

What must an operator do if forecast wind approaches the manufacturer's maximum rated wind speed?

Answer

Recalculate the wind-derated capacity using the manufacturer's reduction schedule and compare it to the actual load weight. If the derated capacity is less than the load, the lift must be delayed.

#61 · Hoisting Calculations

What is center of gravity as it relates to load weight determination?

Answer

The single point where a load's entire weight can be considered concentrated for calculation purposes. Correctly locating it is essential for balanced picks and accurate rigging point selection.

#62 · Hoisting Calculations

In a two-point pick on an unevenly loaded beam, which pick point carries more of the total weight?

Answer

The pick point closer to the load's center of gravity carries the larger share of the total weight. Pick points only share the load equally when both are equidistant from the center of gravity.

#63 · Hoisting Calculations

Why is accurately determining a load's actual weight, including rigging, critical before consulting the load chart?

Answer

The chart's rated capacity must exceed the total suspended weight (load plus all rigging) for the lift to be safe. Underestimating load weight can result in exceeding the crane's true limits without realizing it.

#64 · Hoisting Calculations

What does a load moment indicator (LMI) system do on a modern mobile crane?

Answer

It continuously monitors radius, boom angle, boom length, and load, and warns or locks out the operator before rated capacity is exceeded. It is a safety backup, not a replacement for chart-based lift planning.

#65 · Hoisting Calculations

Why should an operator never rely solely on the LMI system instead of planning the lift with the load chart first?

Answer

The LMI reacts to conditions as they occur and may not account for every rigging deduction or upcoming radius change. Proper lift planning with the chart prevents dangerous surprises rather than reacting to alarms.

#66 · Hoisting Calculations

What happens to rated capacity when outriggers are only partially extended (mid-extension) rather than fully extended?

Answer

Rated capacity is reduced compared to full extension, since a narrower outrigger base brings the tipping fulcrum closer to the crane's centerline. Charts list separate capacity tables for each outrigger extension setting.

#67 · Hoisting Calculations

Why must an operator always match the load chart to the actual outrigger extension being used?

Answer

Using the fully-extended chart while outriggers are only partially extended overstates the true rated capacity. This mismatch is a common and dangerous chart-reading error.

#68 · Hoisting Calculations

What is the boom angle indicator and what does it measure?

Answer

A gauge, often mounted on the boom, that measures the angle the boom centerline makes with horizontal. It gives the operator a quick cross-check against the radius shown on the LMI or chart.

#69 · Hoisting Calculations

Why does telescoping the boom out to a longer length require rechecking the load chart mid-lift?

Answer

Rated capacity is specific to each boom length column on the chart, so extending the boom moves the lift into a different, usually lower, capacity column. Failing to recheck can put the crane over its rated limit.

#70 · Hoisting Calculations

What is the relationship between the number of sheaves in a reeving system and mechanical advantage?

Answer

More sheaves generally allow more parts of line to be reeved, increasing mechanical advantage and reducing tension per part. Each additional part also introduces a small amount of additional friction loss.

#71 · Hoisting Calculations

How do you convert a ground bearing pressure result from psf to psi?

Answer

Divide the psf value by 144, since there are 144 square inches in one square foot. This conversion matters because some soil or mat ratings are given in psi rather than psf.

#72 · Hoisting Calculations

What is the danger of using a smaller outrigger pad than recommended for a given reaction load?

Answer

A smaller pad increases ground bearing pressure for the same load, which can exceed the soil's safe bearing capacity and cause the outrigger to sink or punch through. This can lead to sudden, uncontrolled crane tipping.

#73 · Hoisting Calculations

Why do sling angle factors increase so quickly as the angle drops below 45 degrees?

Answer

Because the factor is based on $1/\sin(\text{angle})$, and sine values shrink rapidly at small angles, causing tension to rise disproportionately. This nonlinear relationship is why even a small angle reduction near the shallow end is risky.

#74 · Hoisting Calculations

What does it mean when a load chart cell is printed in bold or marked with an asterisk?

Answer

It typically indicates that capacity at that cell is structurally limited rather than tipping limited. This tells the operator that adding counterweight will not increase capacity at that specific radius.

#75 · Hoisting Calculations

If a chart shows tipping-limited capacities at long radius, what change would most likely increase available capacity there?

Answer

Adding counterweight or increasing outrigger spread, both of which improve stability against tipping. These remedies only work where tipping, not structural strength, is the governing limit.

#76 · Hoisting Calculations

What role does the crane's own weight (excluding load) play in outrigger reaction loads?

Answer

The crane's own weight is distributed across all four outriggers along with the load's moment effect, so total reaction load is a combination of both, not the load weight alone. This is why outrigger checks must include the full crane weight, not just the suspended load.

#77 · Hoisting Calculations

Why can a load positioned toward one side of the crane cause unequal outrigger reactions?

Answer

The combined center of gravity of the crane, counterweight, and load shifts toward whichever side carries more weight, increasing the reaction on the outriggers nearest that side. Checking only the average reaction across all four outriggers can miss an overloaded individual pad.

#78 · Hoisting Calculations

What is the difference between gross capacity and net capacity on a load chart?

Answer

Gross capacity is the raw figure printed on the chart before any deductions; net capacity is what remains after subtracting hook block, rigging, and jib weight. The actual load must never exceed net capacity.

#79 · Hoisting Calculations

Why is a load chart's rated capacity always based on the crane being level?

Answer

Charts assume the crane is set up level and out-of-level operation is not covered by the standard capacity ratings. Even a small out-of-level condition can significantly reduce actual safe capacity and increase tipping risk.

#80 · Hoisting Calculations

What information does an operator need before selecting a boom length for a planned lift?

Answer

The required operating radius, the load's total weight including rigging, and the quadrant the lift will be made in. All three must be checked together against the chart before committing to a boom length.

#81 · Hoisting Calculations

What is the purpose of a reeving efficiency factor in multi-part line calculations?

Answer

It accounts for friction losses at each sheave the rope passes over, giving a more accurate actual line pull than the simple frictionless load-divided-by-parts formula. Ignoring efficiency can lead to underestimating true rope tension.

#82 · Hoisting Calculations

Why is it unsafe to substitute an over-front chart capacity for an over-side lift at the same radius?

Answer

Quadrant ratings reflect real stability differences specific to that lift direction, and over-front figures are usually higher because that quadrant is more stable. Using the wrong quadrant's number can result in exceeding the crane's true rated capacity.

#83 · Hoisting Calculations

How should an operator handle a lift where wind, quadrant, and rigging deductions all apply at once?

Answer

Each limit must be checked independently and the most restrictive one must govern the decision, not just the first one checked. A lift is only safe when it satisfies every applicable limit simultaneously.

#84 · Hoisting Calculations

What is the pin height used for in boom tip height calculations?

Answer

It is the height of the boom foot pin above ground level, added to the boom length times sine of the angle to get true boom tip height above the ground, not just above the pin. This matters for clearance around structures and power lines.

#85 · Hoisting Calculations

Why does a longer boom sometimes give a higher rated capacity than a shorter boom at the same operating radius?

Answer

Reaching the same radius with a longer boom requires a steeper boom angle, which shifts more of the load's moment closer to the crane's center of rotation, improving both tipping and structural loading. This is why operators sometimes choose a longer boom setup even for a relatively short radius.

#86 · Hoisting Calculations

What should an operator do immediately after any change in radius, quadrant, boom length, or rigging during a lift?

Answer

Recheck the load chart and all applicable deductions and derating factors before continuing the move. Any single changed variable can shift the lift outside of safe rated capacity.

#87 · Hoisting Calculations

What is the typical wind derating pattern used on many mobile crane charts?

Answer

Full capacity is rated up to a threshold wind speed, with a percentage reduction applied for each increment above that threshold, up to a stated maximum operating wind speed. Above the maximum, all lifting must stop.

#88 · Hoisting Calculations

Why is verifying arithmetic critical when working with hoisting calculations?

Answer

A single arithmetic error in radius, sling tension, or ground bearing pressure can lead directly to exceeding rated capacity without any warning signs on site. Hoisting calculations should always be independently double-checked before a lift begins.

#89 · Hoisting Calculations

What does the term rated capacity mean on a mobile crane load chart?

Answer

The maximum load, including all rigging, that the manufacturer certifies the crane can safely lift at a specific boom length, radius, and configuration. It already reflects whichever limit, tipping or structural, is more restrictive at that cell.

#90 · Hoisting Calculations

How does the number of outriggers in use affect ground bearing pressure calculations?

Answer

Ground bearing pressure is calculated per individual outrigger pad, using that pad's own reaction load and area, not an average across all four. Each outrigger must independently meet the soil's safe bearing capacity.

#91 · Hoisting Calculations

What is a spreader bar and why does its weight matter in hoisting calculations?

Answer

A spreader bar is a rigging device that helps control sling angles and distribute load across multiple pick points; its weight adds to the total suspended weight that must fit within net capacity. Forgetting to include it in load weight determination is a common rigging error.

#92 · Hoisting Calculations

Why does a shallow sling angle make it harder to fit rigging within available headroom, and what tradeoff does that create?

Answer

Increasing the sling angle from horizontal (making it steeper) reduces leg tension but requires more vertical headroom above the load. Riggers must balance available clearance against acceptable sling angle when planning the pick.

#93 · Hoisting Calculations

What does it mean if a load chart shows no capacity value at all for a given radius and boom length?

Answer

That configuration is outside the crane's rated working range and lifting is not permitted there, regardless of load weight. Operators must select a boom length and radius combination that has a listed rated capacity.

#94 · Hoisting Calculations

What is the sling leg tension multiplier for a 75 degree sling angle from horizontal?

Answer

The multiplier is approximately 1.035, close to the ideal 90 degree vertical angle. Sling angles at or above 60 degrees are generally considered efficient and low-risk.

#95 · Hoisting Calculations

What is the sling leg tension multiplier at exactly 90 degrees (vertical legs)?

Answer

The multiplier is 1.000, meaning each leg carries exactly its share of the load with no angle penalty. This is the most efficient sling configuration but is not always achievable due to headroom or load geometry.

#96 · Hoisting Calculations

Why do load charts list rated capacity separately for over-front, over-side, and over-rear quadrants?

Answer

Because the crane's stability margin, set by the outrigger footprint and carrier geometry, is not the same in every direction. Using the wrong quadrant column is a common cause of exceeding true rated capacity.

#97 · Hoisting Calculations

Which quadrant typically offers the highest rated capacity on most mobile cranes: front, side, or rear?

Answer

Over-rear typically offers the highest rated capacity, since the crane's counterweight is concentrated there, maximizing the stabilizing moment against tipping. Over-side is typically the most restrictive, since the crane's stability base is narrowest in that direction.

#98 · Hoisting Calculations

What is the difference between a load's actual weight and the rated capacity shown on the chart?

Answer

Actual weight is what the load plus all rigging truly weighs; rated capacity is the maximum the crane is certified to lift at that configuration. The lift is only safe when actual weight is at or below net rated capacity.

#99 · Hoisting Calculations

Why is it important to know a load's weight before selecting sling leg length and angle?

Answer

Sling leg tension depends directly on load weight and angle, so an inaccurate weight estimate can lead to selecting undersized slings or an unsafe angle. Weight should be confirmed from documentation, not estimated visually.

#100 · Hoisting Calculations

What happens to available rated capacity if a crane operates on outriggers that are not fully leveled?

Answer

Available safe capacity is reduced below the chart's rated figure, since charts assume a level set-up. Out-of-level operation increases the risk of tipping even within the printed rated capacity.

#101 · Hoisting Calculations

How does adding counterweight typically affect a tipping-limited lift versus a structurally-limited lift?

Answer

Additional counterweight increases rated capacity where tipping governs, but has no effect where the boom or structure itself is the limiting factor. Knowing which limit applies tells the operator whether more counterweight will actually help.

#102 · Hoisting Calculations

What is the purpose of a load chart's boom angle column when radius is already listed?

Answer

It lets the operator cross-check the boom angle indicator reading against the expected radius for that boom length, catching setup errors. If the indicated angle does not match the chart's expected angle for that radius, something is misconfigured.

#103 · Hoisting Calculations

Why should an operator always use the actual configuration in use (jib stowed vs. erected) when reading the chart?

Answer

Jib-erected and jib-stowed configurations have different weight, length, and moment characteristics, so they use entirely separate chart pages. Reading the wrong chart page can significantly overstate true available capacity.

#104 · Hoisting Calculations

What does mechanical advantage mean in the context of multi-part line reeving?

Answer

It is the ratio by which reeving multiplies the pulling force available at the hook compared to the tension in a single line part. More parts of line means more mechanical advantage but slower load travel speed.

#105 · Hoisting Calculations

Why is ground bearing pressure especially critical on soft or uneven job sites?

Answer

Soft or uneven ground has a lower safe bearing capacity, so the same reaction load that would be safe on solid ground could cause pad sinkage or crane instability. Mats, cribbing, or larger pads are often required on such sites.

#106 · Hoisting Calculations

What information is needed to calculate the tension in a single sling leg of an asymmetric multi-leg rig?

Answer

The share of the total load carried by that leg (based on center of gravity position) and that leg's specific angle from horizontal. Asymmetric rigs require checking each leg individually rather than assuming an even split.

#107 · Hoisting Calculations

Why does a crane's rated capacity generally decrease as boom length increases, even at the same radius?

Answer

A longer boom is a longer lever arm and adds more of its own weight and flexibility to the system, both of which reduce structural and tipping capacity compared to a shorter boom reaching the same radius at a shallower angle.

#108 · Hoisting Calculations

What does the term working load limit (WLL) mean for a sling or rigging component?

Answer

The maximum load a specific piece of rigging is rated to carry under normal conditions, set by the manufacturer with a built-in safety factor. Sling angle tension must never be allowed to exceed the sling's WLL.

#109 · Hoisting Calculations

Why must both crane capacity and rigging capacity be checked independently before a lift?

Answer

Crane capacity depends on radius, boom length, and quadrant, while rigging capacity depends on sling angle and load distribution; a lift within crane capacity can still exceed sling capacity or vice versa. Both checks must pass for the lift to be safe.

#110 · Hoisting Calculations

What is a headache ball and how does its weight factor into hoisting calculations?

Answer

It is a weighted ball used in place of a hook block on some rigs to keep the hoist line taut; like a hook block, its weight must be deducted from gross chart capacity to find net capacity for the load.

#111 · Hoisting Calculations

How can an operator estimate ground bearing pressure before outriggers are even set, during lift planning?

Answer

By estimating the expected reaction load at each outrigger from the planned crane and load configuration, then dividing by the pad size that will be used. This lets mats or cribbing be arranged in advance rather than discovered as inadequate on site.

#112 · Hoisting Calculations

What does it mean if two chart cells for the same radius show a large capacity drop between adjacent boom lengths?

Answer

It usually signals a transition between structural and tipping governing limits, or a change in the crane's configuration such as counterweight requirements. Operators should not assume capacity changes smoothly across every boom length.

#113 · Hoisting Calculations

Why is it good practice to build in a safety margin beyond the exact net capacity when planning a lift?

Answer

Real-world factors like wind gusts, minor weight estimation error, or dynamic loading during hoisting can momentarily exceed the static rated capacity. A margin between planned load and net capacity helps absorb these variables safely.

#114 · Hoisting Calculations

What is dynamic loading and why does it matter even when a static lift plan shows adequate capacity?

Answer

Dynamic loading is the extra force created by sudden starts, stops, or swinging during a lift, which can momentarily exceed the static weight of the load. Smooth, controlled crane operation helps minimize dynamic loading beyond planned capacity.

#115 · Hoisting Calculations

Why do riggers prefer using a spreader beam over a simple two-leg bridle for very wide loads?

Answer

A spreader beam can keep sling legs closer to vertical even on a wide load, avoiding the high tension penalty of a shallow sling angle. This often results in lower sling tension despite the added weight of the beam itself.

#116 · Hoisting Calculations

What should an operator verify about outrigger extension before comparing to any load chart figures?

Answer

That the chart being used matches the actual outrigger extension setting (fully extended, mid-extended, or retracted) in use for that lift. Charts are configuration-specific and mismatching them overstates real capacity.

#117 · Hoisting Calculations

Why does capacity typically improve when a crane's outriggers are extended to their widest setting?

Answer

A wider outrigger base moves the tipping fulcrum farther from the crane's centerline, increasing the stability margin against tipping. This is why fully extended outriggers are used whenever site space allows.

#118 · Hoisting Calculations

What is the relationship between wind speed and dynamic load swing during a lift?

Answer

Higher wind speed increases the tendency of a suspended load to swing, adding unpredictable dynamic forces on top of the load's static weight. This is part of why wind derating becomes stricter as speed increases.

#119 · Hoisting Calculations

Why is understanding both trigonometry and load chart reading essential for hoisting calculations?

Answer

Trigonometry lets an operator predict radius and height from boom angle before consulting the chart, while the chart provides the certified rated capacity for that configuration. Both skills together allow safe, accurate lift planning.

#120 · Hoisting Calculations

What should an operator do if calculated sling leg tension is close to, but just under, the sling's working load limit?

Answer

Treat it as a marginal condition and consider a larger sling, a steeper angle, or a spreader beam rather than proceeding with minimal margin. A tight margin leaves little room for real-world variables like dynamic loading.

#121 · Crane Inspection and Maintenance

What is a daily pre-operational inspection?

Answer

A quick check the operator does before every shift starts. It looks at things like fluid levels, tires or tracks, controls, and obvious damage to catch problems before the crane is used that day.

#122 · Crane Inspection and Maintenance

How many randomly distributed broken wires in one rope lay usually means discard?

Answer

As a general rule, six or more randomly distributed broken wires in one lay of rope means the rope must be taken out of service. A rope lay is the length it takes for one strand to make a full wrap around the rope.

#123 · Crane Inspection and Maintenance

How many broken wires in a single strand within one lay usually means discard?

Answer

Three or more broken wires in one strand within a single lay is enough to discard the rope, even if the total wire count across the whole rope is low. Damage concentrated in one strand is more dangerous than the same damage spread out.

#124 · Crane Inspection and Maintenance

What is a valley break in wire rope?

Answer

A valley break is a broken wire at the point where two strands touch each other inside the rope. It is a sign of internal fatigue and even one valley break is grounds for taking the rope out of service.

#125 · Crane Inspection and Maintenance

What diameter reduction in wire rope usually triggers discard?

Answer

A noticeable reduction in nominal diameter, often around the 5 percent range depending on the standard, means the rope must be discarded. This can happen from internal wear, core loss, or corrosion even when the outside looks fine.

#126 · Crane Inspection and Maintenance

What is birdcaging in a wire rope?

Answer

Birdcaging is when the outer strands separate and bulge away from the rope's core, usually from sudden loss of tension. It permanently disrupts how the rope shares load and is an automatic discard condition.

#127 · Crane Inspection and Maintenance

What is kinking in a wire rope?

Answer

A kink is a permanent bend or twist in the rope caused by improper handling, like pulling rope off a spool the wrong way. A kinked rope must be removed from service because the kink permanently weakens that section.

#128 · Crane Inspection and Maintenance

Why is heat discoloration on a wire rope or steel structure a concern?

Answer

Heat exposure can change the internal properties of steel or rope, making it weaker even if there is no visible bending or breakage. It is treated as a serious concern until an inspection proves the material is still sound.

#129 · Crane Inspection and Maintenance

What should an operator check on a hook during inspection?

Answer

Check the throat opening for spreading, look for twist out of the hook's original plane, check for cracks, and confirm the safety latch closes and holds properly. Any of these defects can mean the hook must be removed from service.

#130 · Crane Inspection and Maintenance

What does an increased hook throat opening indicate?

Answer

It means the hook has been overloaded or stressed beyond its design limit and has started to bend open. Once the spread passes the allowed limit, the hook must be taken out of service.

#131 · Crane Inspection and Maintenance

What is hook twist and why does it matter?

Answer

Hook twist is when the hook has rotated out of its original flat plane, usually from side loading. Excess twist weakens the hook and changes how the load sits in it, so it is checked against an allowable angle limit.

#132 · Crane Inspection and Maintenance

What should be checked on a load block during inspection?

Answer

Check the sheaves for wear or cracks, confirm the swivel bearing turns smoothly without grinding, and inspect the block housing for cracks or bent parts. A rough or noisy swivel bearing needs further inspection before use.

#133 · Crane Inspection and Maintenance

Why does uneven sheave groove wear matter?

Answer

An unevenly worn groove creates an irregular surface that can cause the rope riding in it to wear abnormally, such as flattened strands on one side. Both the sheave and the affected rope section need to be checked and corrected.

#134 · Crane Inspection and Maintenance

What is checked during a hydraulic system inspection?

Answer

Check fluid level and condition, look for leaks at hoses and fittings, watch for cylinder drift under a held load, and confirm pressure gauges read within normal range. Any drift or bypass under load points to a worn internal seal.

#135 · Crane Inspection and Maintenance

What does a bulging hydraulic hose mean, even with no leak visible?

Answer

A bulge means the internal reinforcement layer inside the hose has already failed. The hose can rupture suddenly under pressure, so it must be tagged out and replaced even though it is not currently leaking.

#136 · Crane Inspection and Maintenance

What causes a hydraulic cylinder to drift downward under a held load?

Answer

A worn or leaking internal seal lets fluid bypass from one side of the piston to the other. This slowly lowers the load even though the controls are centered and the pump and fluid level test normal.

#137 · Crane Inspection and Maintenance

What should be checked during a boom or structural inspection?

Answer

Look for dents, bent sections, cracked welds, loose or missing fasteners, and corrosion. Cracks at weld joints are treated as an immediate lockout condition because they can grow under load.

#138 · Crane Inspection and Maintenance

How is a dent in a boom section different from a weld crack?

Answer

A shallow dent without cracking usually needs an engineering evaluation to confirm the section is still strong enough, but it is not always an automatic lockout. A weld crack is a structural discontinuity and requires immediate lockout.

#139 · Crane Inspection and Maintenance

What is a preventive maintenance schedule?

Answer

A planned set of maintenance tasks, like oil sampling, filter changes, and lubrication, performed at set time or hour intervals. It is designed to catch wear before it causes a failure, not just to react after something breaks.

#140 · Crane Inspection and Maintenance

Should a maintenance task be skipped if the equipment sounds and looks fine?

Answer

No. Maintenance intervals are based on elapsed hours or time because internal wear can progress without any noise or visible leak. Scheduled tasks that are overdue should be performed as soon as possible, not skipped.

#141 · Crane Inspection and Maintenance

Why is lubrication important for wire rope?

Answer

Lubrication reduces friction between strands and wires, protects against corrosion, and helps the rope resist internal wear. Poor lubrication speeds up both surface and internal deterioration of the rope.

#142 · Crane Inspection and Maintenance

What should be done with grease points on a crane during preventive maintenance?

Answer

Each grease point should be lubricated on its scheduled interval with the correct grease type until fresh grease is visible. Skipping or over-spacing lubrication intervals speeds up wear on pins, bearings, and sheaves.

#143 · Crane Inspection and Maintenance

What is deficiency reporting?

Answer

The process of formally recording any defect or problem found during an inspection, including its location, severity, and what action was taken. It creates a paper trail so defects are tracked and repaired, not forgotten.

#144 · Crane Inspection and Maintenance

What does lockout mean for a crane with a serious defect?

Answer

Lockout means physically and administratively taking the crane out of service so it cannot be operated until the defect is repaired and verified. It protects workers from using equipment with a known safety hazard.

#145 · Crane Inspection and Maintenance

What is an anti-two-block device?

Answer

A safety device that stops hoisting movement before the load block contacts the boom tip sheaves. If it fails a function test, the crane must be locked out immediately because a critical safeguard is missing.

#146 · Crane Inspection and Maintenance

If an anti-two-block device fails its test and the boom angle indicator is slightly out of calibration, which is the bigger safety concern?

Answer

The failed anti-two-block device is the bigger concern because it removes a safeguard against sudden overload and structural failure. A boom angle indicator that is a few degrees off reduces accuracy but is not an immediate safety device failure.

#147 · Crane Inspection and Maintenance

What is CSA Z150?

Answer

CSA Z150 is the Canadian safety standard for mobile cranes. It covers design, inspection, maintenance, and safe operating requirements that Canadian crane operators and employers must follow.

#148 · Crane Inspection and Maintenance

Why is CSA Z150 awareness important for a hoisting engineer?

Answer

It sets the baseline rules for inspection frequency, maintenance practices, and safe operation across Canada. Knowing it helps an operator recognize when a crane condition falls outside the accepted safety standard.

#149 · Crane Inspection and Maintenance

What is corrosion pitting on wire rope?

Answer

Small pits or craters on the surface of the wire caused by rust and moisture exposure. Scattered pitting without measurable diameter loss is usually a monitoring item, but pitting that reduces diameter or wire strength becomes a discard concern.

#150 · Crane Inspection and Maintenance

What does it mean if wire rope shows flattened strands?

Answer

Flattened strands mean the rope has been under abnormal contact pressure, often from riding on a worn or damaged sheave. It signals a problem that needs correction, not just a normal wear pattern.

#151 · Crane Inspection and Maintenance

What is the difference between random broken wires and concentrated broken wires?

Answer

Random broken wires are spread out across the whole rope and are compared against a whole-rope discard count. Concentrated broken wires are clustered in one strand or one small area, and they reach a discard condition at a much lower count because they weaken one load path severely.

#152 · Crane Inspection and Maintenance

What is core support loss in wire rope?

Answer

Core support loss happens when the rope's inner core deteriorates or shrinks, leaving the outer strands without proper backing. It often shows up only as a diameter reduction, with no visible outer damage.

#153 · Crane Inspection and Maintenance

How often should hydraulic fluid be checked as part of maintenance?

Answer

Hydraulic fluid level and condition should be checked as part of the daily pre-operational inspection, with a full fluid analysis or change done on the manufacturer's scheduled interval.

#154 · Crane Inspection and Maintenance

What does gritty or grinding feel in a swivel bearing indicate?

Answer

It usually means the bearing surfaces are worn or contaminated, even if the block still spins freely by hand. Smooth rotation alone does not confirm the bearing is in good condition, so it needs further inspection.

#155 · Crane Inspection and Maintenance

What is the purpose of an oil sample analysis on a gearbox?

Answer

It checks for metal particles, contamination, and fluid breakdown that indicate internal wear before it causes a failure. It is a preventive tool used even when the gearbox sounds and runs normally.

#156 · Crane Inspection and Maintenance

Why should overdue maintenance be treated as urgent even without symptoms?

Answer

Because internal component wear does not always produce noise, leaks, or visible signs until damage is advanced. Waiting for a symptom before doing overdue maintenance risks a failure that could have been prevented.

#157 · Crane Inspection and Maintenance

What is a weeping leak at a hydraulic fitting?

Answer

A slow, minor fluid leak at a connection point, often from a worn seal or loose fitting. It should be tagged and repaired since it indicates the seal is already compromised and can worsen under pressure and vibration.

#158 · Crane Inspection and Maintenance

Can a bulging hose be temporarily fixed with tape and monitored?

Answer

No. A bulge means internal hose reinforcement has already failed and the hose can rupture without warning under pressure. It must be tagged out and replaced, not patched or monitored.

#159 · Crane Inspection and Maintenance

What is the purpose of a wire rope discard criteria chart?

Answer

It lists the specific conditions, like broken wire counts, diameter loss, kinks, and corrosion, that mean a rope must be taken out of service. It gives inspectors clear, consistent standards instead of relying on guesswork.

#160 · Crane Inspection and Maintenance

What should an operator do if a defect is found during the daily inspection?

Answer

Stop and evaluate the severity, report the deficiency following the proper reporting process, and lock out the crane if the defect affects safe operation. Minor items can be logged for scheduled repair if they do not create an immediate hazard.

#161 · Crane Inspection and Maintenance

Why is it unsafe to keep operating with a slightly bent hook?

Answer

A bent or spread hook has already been stressed beyond its normal limit and is weaker than a hook in its original shape. Continuing to use it risks the hook opening further or failing under load.

#162 · Crane Inspection and Maintenance

What is the danger of ignoring surface rust on a boom without pitting?

Answer

Light surface rust without pitting is usually a minor maintenance item, but it should still be logged and monitored because untreated corrosion can progress to pitting and structural weakening over time.

#163 · Crane Inspection and Maintenance

Why can a rope be discarded even if the total broken wire count seems low?

Answer

Because where the broken wires are located matters more than the total count. A small number of breaks concentrated in one strand or at a valley point can trigger discard faster than a larger number spread across the whole rope.

#164 · Crane Inspection and Maintenance

What is crushing damage on a wire rope?

Answer

Crushing happens when the rope is flattened or deformed, often from being run over, over-wound, or pinched. It disrupts the internal structure of the rope and is a discard condition.

#165 · Crane Inspection and Maintenance

Why should a technician never repair a cracked boom weld by field welding without evaluation?

Answer

Because boom steel has specific strength and heat treatment requirements, and an unevaluated field weld can introduce new weaknesses or hidden defects. Structural repairs need proper engineering evaluation and qualified repair procedures first.

#166 · Crane Inspection and Maintenance

What is the purpose of checking fasteners during structural inspection?

Answer

Loose or missing bolts and pins can allow structural components to shift or separate under load. Checking fasteners catches this before it becomes a serious structural failure.

#167 · Crane Inspection and Maintenance

How does contamination affect hydraulic fluid performance?

Answer

Contaminated fluid, from dirt, water, or worn metal particles, can cause pump cavitation, valve sticking, and accelerated wear throughout the hydraulic system. Regular fluid checks help catch contamination early.

#168 · Crane Inspection and Maintenance

What does a slow pressure loss in a hydraulic system without visible leaks suggest?

Answer

It suggests internal bypass, most often from a worn seal inside a cylinder or valve, letting fluid move where it should not. This is different from an external leak, which would show visible fluid.

#169 · Crane Inspection and Maintenance

Why do inspection standards separate hook throat opening and hook twist as different criteria?

Answer

Because they are independent types of damage caused by different loading conditions, straight overload versus side loading. A hook can fail one criterion without failing the other, so both must be checked separately.

#170 · Crane Inspection and Maintenance

What is the role of a boom angle indicator?

Answer

It shows the operator the current angle of the boom, which is used with the load chart to determine the crane's safe lifting capacity at that position. An inaccurate reading can lead to using the wrong capacity limit.

#171 · Crane Inspection and Maintenance

Why is an inaccurate boom angle indicator dangerous but not always an immediate lockout?

Answer

Because it reduces the operator's accuracy in reading the load chart, but the crane can often still be operated safely using backup methods until it is recalibrated. A completely failed safety device, like anti-two-block, is treated more urgently.

#172 · Crane Inspection and Maintenance

What is the general idea behind discard criteria for cranes and rigging equipment?

Answer

Discard criteria set measurable limits, like wire counts, diameter loss, or angle of bend, so that equipment is removed from service before it becomes unsafe rather than after it fails.

#173 · Crane Inspection and Maintenance

Why should inspection findings be recorded even when the equipment stays in service?

Answer

Recording minor findings creates a history that helps track how fast wear is progressing. It also proves the inspection was actually done and provides evidence for future maintenance decisions.

#174 · Crane Inspection and Maintenance

What does it mean if two separate defects are found during one inspection?

Answer

Each defect should be evaluated on its own severity, not averaged together. A minor defect does not become urgent just because a serious one was also found, and a serious one still requires immediate action even if the other is minor.

#175 · Crane Inspection and Maintenance

What is a function test in the context of safety devices?

Answer

A function test actively operates a safety device, like an anti-two-block system, to confirm it actually stops the movement it is designed to prevent, rather than just visually checking that it is installed.

#176 · Crane Inspection and Maintenance

Why is visual inspection alone sometimes not enough for wire rope?

Answer

Some serious defects, like internal core loss or valley breaks, are not visible from the outside. This is why diameter measurement and careful wire-by-wire checking are also required, not just a visual scan.

#177 · Crane Inspection and Maintenance

What should be done if a maintenance interval and an inspection finding conflict, like equipment overdue but appearing fine?

Answer

The scheduled maintenance interval takes priority because it is based on known wear patterns over time, not on how the equipment currently looks or sounds. Overdue maintenance should be completed as soon as possible.

#178 · Crane Inspection and Maintenance

What is the purpose of comparing measured rope diameter to the nominal diameter?

Answer

Comparing the current measured diameter to the rope's original nominal diameter is how inspectors detect internal wear or core loss that would not otherwise be visible from the outside.

#179 · Crane Inspection and Maintenance

Why does a crack always outrank surface rust in urgency during a structural inspection?

Answer

A crack is an active structural discontinuity that can grow under load and lead to sudden failure, while surface rust without pitting develops slowly and can be addressed on a normal maintenance schedule.

#180 · Crane Inspection and Maintenance

What is the danger of operating a crane with a known but unreported deficiency?

Answer

Without a report, the deficiency will not be tracked or scheduled for repair, and other operators using the crane will not know about the hazard. Reporting is what turns a known problem into a managed one.

#181 · Crane Inspection and Maintenance

Why is it important to check both the sheave and the rope together during inspection?

Answer

Damage often has a shared cause. A worn sheave can cause abnormal rope wear on the same side, so fixing only one component without checking the other can leave the root cause in place.

#182 · Crane Inspection and Maintenance

What does reducing rated capacity instead of discarding a rope achieve, and when is it appropriate?

Answer

Reducing rated capacity is not a substitute for meeting discard criteria. Once a rope reaches a defined discard condition, like a valley break or excessive broken wires, it must be removed from service, not simply derated.

#183 · Crane Inspection and Maintenance

Why should a technician not assume a component is fine just because it passed inspection last time?

Answer

Wear and damage develop over time and can appear between inspections. Each inspection needs to independently evaluate current condition rather than relying on a past result.

#184 · Crane Inspection and Maintenance

What is the overall goal of the daily inspection, preventive maintenance, and deficiency reporting system together?

Answer

Together they catch problems early, through daily checks, scheduled service, and documented reporting, so defects are found and corrected before they cause an accident rather than after.

#185 · Crane Inspection and Maintenance

Why do CSA Z150 style standards separate criteria for wire rope, hooks, hydraulics, and structure instead of using one general rule?

Answer

Each component fails in different ways under different stresses, so a single general rule would either be too strict for some parts or too loose for others. Separate, specific criteria give inspectors accurate, component-appropriate limits.

#186 · Rigging

What does WLL stand for in rigging?

Answer

WLL stands for working load limit. It is the maximum weight a sling or piece of hardware can safely carry under normal, straight, vertical use. It is stamped or tagged by the manufacturer and must never be exceeded.

#187 · Rigging

Why is wire rope sensitive to fatigue over time?

Answer

Wire rope is made of many twisted steel wires, and repeated bending and loading gradually breaks individual wires. Once enough wires break, the sling loses rated strength even if it still looks intact from a distance. This is why regular inspection for broken wires matters so much.

#188 · Rigging

Why is chain a good choice for hot or sharp loads?

Answer

Chain resists heat and sharp edges far better than synthetic materials, and it also stretches somewhat before it fails. This stretch can act as a visible early warning sign. Chain is heavier for its strength than wire rope or synthetics, but it holds up in harsh conditions.

#189 · Rigging

What is the main weakness of synthetic web and round slings?

Answer

Synthetic slings made of nylon or polyester are vulnerable to heat and chemical exposure. Damage from these sources can weaken the sling with no visible change in appearance. This makes checking exposure history just as important as a visual inspection.

#190 · Rigging

Describe a vertical hitch.

Answer

A vertical hitch attaches the sling straight from the hook to a single point on the load with no wrap around the load. It gives the sling its full rated capacity but offers no control against rotating or sliding. It is used when the load has a secure engineered lift point.

#191 · Rigging

Describe a choker hitch and its main benefit.

Answer

A choker hitch wraps around the load and passes one end through a loop formed by the other, cinching tight as tension increases. This grip works well on round or bundled loads because it tightens itself. The bend at the choke point reduces the sling's rated capacity compared to vertical.

#192 · Rigging

Describe a basket hitch and its effect on capacity.

Answer

A basket hitch cradles the load with both ends of the sling going up to the hook, effectively using two load bearing legs instead of one. With both legs vertical, this roughly doubles the vertical rated capacity. Spreading the legs away from vertical reduces that doubled capacity again.

#193 · Rigging

What is the sine value at 90 degrees from horizontal, and what does it mean for tension?

Answer

At 90 degrees, straight up, the sine value is 1. This means the tension in the sling leg equals exactly its share of the load weight, with no increase from angle at all.

#194 · Rigging

What is the sine value at 60 degrees from horizontal, and what tension multiplier does it create?

Answer

The sine of 60 degrees is about 0.866. Dividing the load share by this value creates a tension multiplier of about 1.15 times the vertical value.

#195 · Rigging

What is the sine value at 45 degrees from horizontal, and what tension multiplier does it create?

Answer

The sine of 45 degrees is about 0.707. Dividing the load share by this value creates a tension multiplier of about 1.41 times the vertical value.

#196 · Rigging

What is the sine value at 30 degrees from horizontal, and what tension multiplier does it create?

Answer

The sine of 30 degrees is 0.5. Dividing the load share by this value doubles the tension compared to the vertical value, making 30 degrees a common minimum threshold in rigging rules.

#197 · Rigging

What is the basic formula for tension in a sling leg?

Answer

Tension equals the load share carried by that leg divided by the sine of the angle from horizontal. As the angle gets smaller, moving further from vertical, the sine value drops and tension climbs.

#198 · Rigging

What sling angle is generally treated as a red flag requiring a revised rigging plan?

Answer

An angle under 30 degrees from horizontal is treated as a red flag. At that point tension is climbing sharply for every additional degree lost, so companies often require longer slings or a spreader bar instead.

#199 · Rigging

What is a shackle's rated capacity based on regarding pull direction?

Answer

A shackle's rated capacity assumes the pull is in line with the pin, not at a sharp side angle. Loading a shackle sharply to the side reduces its safe capacity below the stamped rating.

#200 · Rigging

What are the two common shackle pin styles?

Answer

The two common styles are screw pin, tightened by hand turning, and bolt type, which uses a bolt with a washer, nut, or cotter pin. Bolt type shackles resist loosening from vibration better than screw pin shackles.

#201 · Rigging

What must a rigging hook have to be considered safe for use?

Answer

A rigging hook must have a functioning safety latch that closes off the throat opening so a sling cannot slip out under slack conditions. Any visible bending, twisting, or an oversized throat opening means the hook must be tagged out immediately.

#202 · Rigging

What is the purpose of a spreader bar?

Answer

A spreader bar holds the top attachment points of a sling apart so the legs running down to a wide load stay closer to vertical. Keeping legs closer to vertical keeps tension lower even on a wide load.

#203 · Rigging

What is the purpose of an equalizer beam?

Answer

An equalizer beam balances the share of load carried by each sling leg. This prevents an uneven or off center load from overloading one leg while another leg sits nearly slack.

#204 · Rigging

What is a master link, and why does its rating matter so much?

Answer

A master link connects every leg of a multi leg sling assembly to a single point on the hook. Because every leg's tension passes through this one link, its rated capacity must meet or exceed the combined pull of every attached leg.

#205 · Rigging

For a four leg bridle sling, how many legs does standard safe practice assume are truly sharing the load?

Answer

Standard practice conservatively assumes only three of the four legs are sharing the load evenly. This accounts for slight length differences and hardware variations that almost always mean one leg carries less than a perfectly even share.

#206 · Rigging

For a three leg bridle sling, how many legs does standard safe practice assume are truly sharing the load?

Answer

Standard practice conservatively assumes only two of the three legs are sharing the load evenly. This mirrors the same conservative approach used for a four leg bridle, just scaled down by one leg.

#207 · Rigging

What is the formula for total safe capacity of a four leg bridle using the three leg assumption?

Answer

Total safe capacity equals one leg's vertical rating times the sine of the angle from horizontal, times three legs. For example, a 2,000 pound leg at 60 degrees gives 3 times 2,000 times 0.866, about 5,196 pounds.

#208 · Rigging

Why can you not simply add up all the tag ratings on a multi leg sling to find total capacity?

Answer

Real world sling assemblies rarely share the load perfectly evenly, and angle reduces how much of each leg's rated strength is usable for a straight vertical lift. Adding tag ratings directly ignores both of these factors and overstates the safe capacity.

#209 · Rigging

What is a tagline, and what does it carry?

Answer

A tagline is a separate rope, unrelated to the lifting slings, attached to a load and held by a ground worker to control rotation and swing. It carries no lifting weight at all.

#210 · Rigging

When are taglines most important during a lift?

Answer

Taglines matter most on long, flat, or irregularly shaped loads, and on windy days when a suspended load naturally wants to spin or drift. A large flat panel can catch wind like a sail without one.

#211 · Rigging

Where should a worker holding a tagline avoid standing?

Answer

The worker should stay clear of the area directly under the load and out from between the load and any fixed structure. A sudden swing in either of those spots could pin or crush the worker.

#212 · Rigging

Can a tagline be used to physically stop a heavy swinging load?

Answer

No. A tagline is meant to guide the load from a safe distance, not to physically restrain a load that is actively swinging with real momentum. Trying to muscle a heavy swinging load to a stop is a losing fight and a serious injury risk.

#213 · Rigging

What is a below-the-hook lifting device?

Answer

A below-the-hook lifting device is any piece of rigging hardware, other than a simple sling, engineered for a specific lifting task. Examples include lifting beams, coil hooks, plate clamps, and vacuum lifters.

#214 · Rigging

What is a plate clamp used for, and what is one key limitation?

Answer

A plate clamp is used to grip and lift flat steel plate. It has a minimum plate thickness it can safely hold, and clamping a thinner plate can allow the clamp to slip free under load.

#215 · Rigging

What kind of load is a vacuum lifter best suited for?

Answer

A vacuum lifter is used for smooth, sealed panel material such as glass or sheet metal. It creates suction against the surface, so a rough or uneven surface would break the seal.

#216 · Rigging

What kind of load is a coil hook designed for?

Answer

A coil hook is shaped specifically to lift cylindrical steel coils. Using it outside that intended purpose is treated the same as using rigging equipment beyond its rated capacity.

#217 · Rigging

Does a below-the-hook device get inspected as part of the crane's own inspection?

Answer

No. A below-the-hook device needs its own periodic load test and written inspection record, separate from the inspection done on the crane itself. It cannot rely on a recent crane inspection to cover it.

#218 · Rigging

What should a rigger look for when inspecting wire rope slings?

Answer

Look for broken wires, especially clustered broken wires concentrated in one small area, along with kinks, crushing, corrosion, and heat discoloration. Clustered broken wires are considered a serious localized defect.

#219 · Rigging

What does a missing or unreadable sling tag mean for its use?

Answer

A sling with a missing, torn off, or unreadable tag must be treated as unidentified and removed from service immediately. Without the tag, nobody can confirm the sling's rated capacity.

#220 · Rigging

What should a rigger look for when inspecting chain slings?

Answer

Look for stretched or bent links, gouges, nicks, and any unexpected twist in the chain. A quick check compares the current chain length to its stamped reference length, since chain measurably stretches before it breaks.

#221 · Rigging

What should a rigger look for when inspecting synthetic web and round slings?

Answer

Look for cuts, snags, punctures, melted or glazed fibers from heat, and exposed core fibers where a round sling's outer cover has worn through. Chemical staining should be treated as a possible strength loss even without visible damage.

#222 · Rigging

What should a rigger look for when inspecting hardware like shackles, hooks, and master links?

Answer

Look for bending, stretching, nicks, gouges, corrosion pitting, and any sign a pin no longer seats fully or threads smoothly. A pin that must be forced, or a hook throat opened wider than its stamped limit, both mean the part must be tagged out.

#223 · Rigging

Why does a two leg bridle sling at a 30 degree angle carry more risk than one at 90 degrees?

Answer

At 30 degrees, the sine value is only 0.5, doubling the tension compared to the vertical share. At 90 degrees the sine value is 1, so there is no tension increase at all. The shallower the angle, the higher the risk of exceeding the sling's rating.

#224 · Rigging

What does side loading a shackle mean, and why is it a problem?

Answer

Side loading means pulling a shackle at a sharp angle away from being in line with the pin. This reduces the shackle's safe capacity below its stamped rating, since the rating assumes an in-line pull.

#225 · Rigging

Why is a bent hook considered dangerous even if it still functions?

Answer

A bent or twisted hook is evidence the hook was already overloaded at some point in its history. Even if it still appears to function, its structural integrity is compromised and it must be tagged out.

#226 · Rigging

What is the difference between a spreader bar and an equalizer beam?

Answer

A spreader bar keeps sling legs closer to vertical by holding attachment points apart, lowering tension on a wide load. An equalizer beam instead balances the share of load carried by each leg, addressing uneven or off center loads.

#227 · Rigging

Why does a smooth round pipe need a choker hitch instead of a vertical hitch?

Answer

A vertical hitch has no wrap and lets a round load roll free out of the sling. A choker hitch cinches around the pipe as tension increases, gripping the surface and resisting rolling.

#228 · Rigging

What percentage rated capacity does a choker hitch commonly provide compared to vertical, and why is it reduced?

Answer

A choker hitch commonly provides about 80 percent of the vertical rated capacity when the choke angle is wide. The bend at the choke point reduces the sling's strength, and a tighter choke angle reduces it even further.

#229 · Rigging

What real world factor can cause one leg of a bridle sling to carry more tension than the simple math predicts?

Answer

An off center center of gravity on the load can cause the leg closer to that center of gravity to carry more than its calculated even share. This means angle math alone is not always enough; load balance matters too.

#230 · Rigging

What is the main hazard of lifting a long flat panel on a windy day without a tagline?

Answer

The panel can catch wind like a sail and swing unpredictably close to workers, structures, or the crane itself. A tagline gives a ground worker the ability to steer and steady the panel from a safe distance.

#231 · Rigging

Why is it unsafe to use a below-the-hook device outside its intended load shape?

Answer

Each below-the-hook device is engineered around one narrow task, such as gripping a specific plate thickness or coil shape. Using it outside that purpose is treated the same as exceeding its rated capacity and risks the load slipping free.

#232 · Rigging

What is the practical difference between breaking strength and working load limit?

Answer

Breaking strength is the point at which a sling actually fails, while working load limit is the safe maximum for normal use, set well below breaking strength. Riggers must always work to the WLL, never toward the breaking strength.

#233 · Rigging

Why does wire rope transmit more shock into a load than a synthetic sling?

Answer

Wire rope has almost no stretch, so it transmits shock loading directly into the load and crane structure if the load is snatched off the ground quickly. Synthetic slings have more give, which absorbs some of that shock.

#234 · Rigging

Why are round slings harder to visually inspect than flat web slings?

Answer

A round sling's outer jacket hides the load bearing core fibers completely, so damage to the core can be invisible from outside. A flat web sling shows surface wear more directly, since its load bearing fibers are more exposed to view.

#235 · Rigging

What is the accepted safe formula for calculating multi-leg sling capacity at an angle?

Answer

Take the vertical rated capacity of one leg, multiply by the sine of the angle from horizontal, then multiply by the conservative number of legs assumed to share the load. This avoids the unsafe shortcut of simply adding every tag rating together.

#236 · Rigging

Why might uneven leg angles in a multi-leg sling be more dangerous than equal angles?

Answer

Uneven angles can push one leg's actual tension well above what a simple average calculation would suggest. Lift plans with uneven angles or an uncertain center of gravity need specific engineering tables rather than a rough estimate.

#237 · Rigging

What is a common site minimum sling angle rule, and why is it set that low and no lower?

Answer

Many sites require a minimum of 45 to 60 degrees from horizontal, with anything under 30 degrees treated as a red flag. Below that threshold, the tension multiplier climbs quickly for each additional degree lost, sharply raising failure risk.

#238 · Rigging

What should a rigger check about a shackle's pin during inspection?

Answer

Check that the pin seats fully and threads smoothly without being forced. A pin that must be forced or does not thread smoothly signals damage and means the shackle must be removed from service.

#239 · Rigging

Why is comparing current chain length to its stamped reference length a useful inspection technique?

Answer

Alloy chain measurably stretches before it fails, so a length longer than the stamped reference length is a warning sign. This gives an inspector a quantifiable way to catch a chain approaching failure before it actually breaks.

#240 · Rigging

What is the risk of using a plate clamp on a plate thinner than its rated minimum thickness?

Answer

The clamp may not grip the thin plate securely and can slip free under load. Plate clamps are engineered around a specific thickness range, so matching the actual plate thickness to the clamp's rating is essential.

#241 · Rigging

Why does a basket hitch lose some of its doubled capacity as the legs spread apart?

Answer

Once the legs move away from vertical, the same sine-based angle reduction that applies to any sling leg applies here too. The angle reduction multiplies against the doubled base capacity, lowering the usable total as the spread increases.

#242 · Rigging

What is the difference in exam question type between the knowledge/recall tier and the critical thinking tier for rigging?

Answer

Knowledge/recall tier questions test straightforward facts and definitions, such as naming a hitch type. Critical thinking tier questions require combining multiple factors, such as angle, hitch type, and number of legs, to judge the safest configuration.

#243 · Rigging

What is the danger of clustered broken wires compared to isolated broken wires spread along a wire rope sling?

Answer

Clustered broken wires concentrated in one small area represent a serious localized loss of strength and require immediate removal from service. Isolated wires spread evenly along the full length are a lower concern, though still tracked during inspection.

#244 · Rigging

Why should a rigger never estimate a sling's rated capacity just from its size?

Answer

Rated capacity depends on the specific construction, material, and manufacturer rating of that exact sling, not just its visible size. Estimating from size alone can significantly overstate or understate the true safe working load.

#245 · Rigging

What is the overall goal of combining hitch selection, angle control, and hardware inspection in a rigging plan?

Answer

Together these choices ensure the load is securely gripped, that leg tension stays within rated limits, and that no damaged component is relied upon during the lift. Skipping any one of these areas can leave a serious gap even if the others are handled correctly.

#246 · Lift Planning and Site Setup

What is a site survey in crane lift planning?

Answer

A site survey is a walk-through of the work area before the crane arrives. It checks the ground, slopes, buried and overhead utilities, and access routes. This information decides where the crane can safely be set up.

#247 · Lift Planning and Site Setup

What is ground bearing pressure?

Answer

Ground bearing pressure is the force per square foot or square meter that a crane's outriggers or tracks push into the soil. It equals the load divided by the contact area. If it is too high for the soil, the crane can sink or tip.

#248 · Lift Planning and Site Setup

What do outrigger mats do?

Answer

Outrigger mats spread the crane's point load over a wider area of ground. This lowers the ground bearing pressure so it stays within what the soil can safely hold. Mat size is chosen based on the load and the soil rating.

#249 · Lift Planning and Site Setup

What is cribbing used for?

Answer

Cribbing is stacked timber or blocks placed under outrigger mats on soft or uneven ground. It adds height and spreads the load even further than a mat alone. It is often stacked in a crisscross pattern for strength.

#250 · Lift Planning and Site Setup

Why must a crane be level before lifting?

Answer

Load charts assume the crane is level within a small tolerance. Being out of level reduces the true safe capacity and can cause the boom to bend sideways. Operators must check the cab level indicator before every lift.

#251 · Lift Planning and Site Setup

What device shows if a crane is level?

Answer

A bubble level or digital level indicator in the operator's cab shows the out-of-level condition. The reading must stay within the manufacturer's stated tolerance. This is checked before boom-up and monitored during the lift.

#252 · Lift Planning and Site Setup

What is minimum approach distance?

Answer

Minimum approach distance is the closest a crane, boom, or load can legally get to an energized power line. The required distance grows larger as the line's voltage increases. It must be confirmed from a clearance table before working near any line.

#253 · Lift Planning and Site Setup

What minimum approach distance is required in Ontario for lines up to 750 volts?

Answer

Ontario construction regulation sets a minimum approach distance of 1.0 meter for energized lines rated up to 750 volts. This is the smallest distance band on the table. Higher voltage lines require significantly more clearance.

#254 · Lift Planning and Site Setup

What minimum approach distance is required in Ontario for lines over 750 volts up to 150 kV?

Answer

Ontario construction regulation sets a minimum approach distance of 3.0 meters for lines in this voltage band. This applies to many common distribution and sub-transmission lines. The exact figure should always be confirmed against the current regulation before working near power lines.

#255 · Lift Planning and Site Setup

What minimum approach distance is required in Ontario for lines over 150 kV up to 250 kV?

Answer

Ontario construction regulation sets a minimum approach distance of 4.5 meters for this voltage band. As voltage increases, so does the required clearance. This range covers many higher-voltage transmission lines.

#256 · Lift Planning and Site Setup

What minimum approach distance is required in Ontario for lines over 250 kV?

Answer

Ontario construction regulation sets a minimum approach distance of 6.0 meters for lines above 250 kV. This applies to major transmission corridors. Crews must confirm the exact voltage before setting up nearby.

#257 · Lift Planning and Site Setup

What should a crew do if a line's voltage is unknown?

Answer

If the exact voltage cannot be confirmed, treat the line as the highest voltage reasonably possible until the utility verifies it. Using a guess or the lowest distance is not safe. Delay the lift if confirmation cannot be obtained.

#258 · Lift Planning and Site Setup

What is a critical lift?

Answer

A critical lift is one that exceeds a set percentage of the crane's rated capacity, involves lifting personnel, uses two cranes together, or has other elevated risk factors. It requires a detailed written plan reviewed before work starts. The threshold percentage is set by company policy or regulation.

#259 · Lift Planning and Site Setup

What load information must a lift plan document?

Answer

A lift plan must document the load's confirmed weight, its center of gravity, and its dimensions. This information is used to select the right rigging and crane setup. Estimating these figures by eye is not acceptable.

#260 · Lift Planning and Site Setup

How should a load's weight be verified?

Answer

Load weight should come from a certified scale ticket, stamped engineering drawing, or another reliable documented source. A shipping label alone is not considered verified. Using an inaccurate weight undermines every other calculation in the plan.

#261 · Lift Planning and Site Setup

What is the universal hand signal for emergency stop?

Answer

Both arms are extended horizontally with palms down and swung back and forth. This signal overrides any other instruction being given at that moment. Every crane operator must obey it immediately, no matter who gives it.

#262 · Lift Planning and Site Setup

What must happen before radio becomes the main lift communication method?

Answer

The crew must assign a dedicated channel and perform a radio check confirming clear transmission and reception. A backup method, such as hand signals, should also be agreed on. Skipping this check risks a missed instruction during the lift.

#263 · Lift Planning and Site Setup

What is an exclusion zone?

Answer

An exclusion zone is a barricaded area around the crane's full swing radius, including the counterweight. It keeps workers, vehicles, and pedestrians clear of the rotating superstructure and any suspended load. It must be sized to the crane's actual maximum working radius, not just its resting footprint.

#264 · Lift Planning and Site Setup

What general wind speed often triggers extra caution or suspension of lifts?

Answer

Around 32 km/h, or about 20 mph, sustained wind is a commonly used general caution threshold. The crane's own load chart or manufacturer manual may set a stricter limit. Large, flat loads often require a much lower limit.

#265 · Lift Planning and Site Setup

Why do large flat loads need a lower wind limit?

Answer

Flat loads like wall panels present a large surface area to the wind, called sail area. This makes them catch more wind force than compact loads. The lift plan should set a load-specific wind limit rather than relying on a general figure.

#266 · Lift Planning and Site Setup

What should happen when lightning is detected near an active lift site?

Answer

Crane operations should be suspended immediately and personnel moved away from the crane and any grounded rigging. The crane acts as a tall conductive path during a storm. Work should not resume until a defined waiting period has passed after the last strike.

#267 · Lift Planning and Site Setup

Who is a signal person and why are they used?

Answer

A signal person is a qualified individual who relays directional, hoist, and stop signals to the operator. They are used when the operator cannot maintain a clear view of the load or hazard area. Only one designated signal person should give active signals at a time.

#268 · Lift Planning and Site Setup

What is a utility locate or notification?

Answer

A utility locate or notification is a request sent to a utility company before crane work occurs near their lines or buried infrastructure. It lets the utility advise on de-energization, line covers, or confirmed clearance. Some jurisdictions require a set number of days' notice.

#269 · Lift Planning and Site Setup

What is the purpose of a permit for crane work?

Answer

A permit authorizes crane work near roadways, utilities, or other regulated areas and may include specific conditions like time windows. Work should not begin until the permit is confirmed in hand. Ignoring permit conditions can lead to noncompliance or a stop-work order.

#270 · Lift Planning and Site Setup

What is a trial lift?

Answer

A trial lift raises the load a short distance off the ground before the full lift proceeds. It verifies the rigging is secure, the load is balanced, and the brakes hold. It gives the crew a chance to catch a problem while the load is still low.

#271 · Lift Planning and Site Setup

Why does sling angle matter in lift planning?

Answer

A shallow sling angle significantly increases the tension in each sling leg compared to a steeper angle. The rigging capacity must be recalculated for the actual angle being used. This applies to any multi-point pick, not just specific configurations.

#272 · Lift Planning and Site Setup

What does a soil bearing capacity rating tell a crew?

Answer

It states how much load per square foot or square meter the ground can support before it fails or settles. This figure is compared against the calculated ground bearing pressure from the crane's setup. Disturbed or fill soil often needs a geotechnical check before this figure can be trusted.

#273 · Lift Planning and Site Setup

Why must the crane's travel route be planned in advance?

Answer

The travel route is checked to confirm the ground can bear the crane's weight and to identify overhead or underground hazards along the way. An unplanned route can lead the crane over an unmarked utility or soft ground. This matters most for crawler cranes traveling under load.

#274 · Lift Planning and Site Setup

How does fog or heavy rain affect a lift?

Answer

Reduced visibility can prevent the operator or signal person from seeing the load, landing zone, or hazards clearly. The lift plan should set a minimum visibility standard for proceeding. When visibility drops too low, the lift should be delayed or extra spotters and lighting added.

#275 · Lift Planning and Site Setup

What is a tandem lift and what extra planning does it need?

Answer

A tandem lift uses two cranes together to lift one load. It needs a documented weight distribution between both cranes and a single lead signal person coordinating both operators. Conflicting signals or uneven load sharing are major risks unique to tandem lifts.

#276 · Lift Planning and Site Setup

Why should mat size be recalculated for each outrigger separately?

Answer

Different outriggers can carry different reaction loads depending on load position and boom orientation. Soil conditions can also vary across a single site. Using one uniform mat size for every position can leave the most heavily loaded outrigger under-supported.

#277 · Lift Planning and Site Setup

What should be done if ground bearing pressure calculates right at the soil's maximum rating with no margin?

Answer

Best practice is to increase mat size or add cribbing to build in a safety margin below the soil's maximum rating. Being exactly at the limit leaves no room for changing conditions, like rain. Soil ratings themselves also carry some uncertainty.

#278 · Lift Planning and Site Setup

How should a lift plan handle background noise that could interfere with communication?

Answer

The plan should combine a dedicated radio channel with confirmed standard hand signals as a backup. Shouted instructions are not a reliable or standardized method in loud conditions. Both channels being available reduces the risk of a missed signal.

#279 · Lift Planning and Site Setup

What extra safety margin often applies to lifts involving a personnel platform?

Answer

Personnel platform lifts typically require a further reduced percentage of the crane's rated capacity compared to standard critical lifts. They also usually need a secondary means of support or fall protection. This reflects the direct risk to workers being lifted.

#280 · Lift Planning and Site Setup

What should the crew do if two sources give conflicting minimum approach distances for the same line?

Answer

Apply the more conservative, larger distance and confirm the correct figure with the utility or the governing regulation. Choosing the smaller figure to save space increases risk. Averaging the two figures is not a valid method.

#281 · Lift Planning and Site Setup

Why does a lift plan specify an exact swing path rather than letting the operator choose?

Answer

The specified path has been checked against known hazards like power lines, structures, and the exclusion zone boundary. A more convenient route might not maintain the same clearances. Once finalized, the documented path becomes the required route.

#282 · Lift Planning and Site Setup

What must be checked if the actual crane setup does not match the load chart on file?

Answer

The physical setup must be corrected, or the crew must select the load chart page that matches the actual configuration, such as counterweight and boom length. Using a mismatched chart risks relying on the wrong rated capacity. Load charts are precise and specific, not general guides.

#283 · Lift Planning and Site Setup

Who has the authority to call an immediate stop during a lift?

Answer

Any crew member who observes a hazard has the authority and responsibility to call an immediate stop. Limiting this to only the operator ignores the value of multiple people watching different aspects of the lift. Delays for outside confirmation are not acceptable during an active hazard.

#284 · Lift Planning and Site Setup

What should happen if a designated signal person becomes unavailable on lift day?

Answer

A new qualified signal person must be identified and fully briefed on the lift plan before any lift movement begins. Allowing an untrained worker to give informal signals creates confusion risk. The lift plan documentation should be updated to reflect who is actually performing the role.

#285 · Lift Planning and Site Setup

Why is a firm-looking surface after heavy rain not always a reliable sign of safe ground?

Answer

Heavy rain can saturate soil beneath a surface that still looks and feels firm on top. The crew should investigate subsurface conditions through probing or a geotechnical check. Increased matting can also help account for reduced bearing capacity.

#286 · Lift Planning and Site Setup

What happens to a load's required lifting configuration if its actual weight comes in heavier than originally planned?

Answer

The lift plan must be revised and re-approved using the new confirmed weight before the lift proceeds. Capacity, rigging, and ground bearing calculations were all based on the original figure. A prior sign-off does not remain valid once a key input changes.

#287 · Lift Planning and Site Setup

What should govern a wind decision when gusts are higher than the sustained wind reading?

Answer

The gust reading should be treated as the controlling factor if it exceeds the plan's stated limit. A sudden gust can occur unpredictably and swing a wind-sensitive load. Relying only on the sustained average understates the real peak force involved.

#288 · Lift Planning and Site Setup

Why should wind readings be taken at boom tip height rather than only at ground level?

Answer

Wind speed generally increases with height, and the load and boom tip are exposed to that higher wind. A ground-level reading alone can understate the actual force on the load. The boom tip reading should govern the lift decision when it differs from ground level.

#289 · Lift Planning and Site Setup

What must a crew confirm about a temporary obstruction placed near the crane after the swing path was first checked?

Answer

The swing path must be reassessed against the new obstruction, and the plan revised or the obstruction relocated before lifting. Relying on the original assessment ignores a real change in site conditions. Estimating a path around it without updating the plan removes the documented verification.

#290 · Lift Planning and Site Setup

What is the concern if a lift plan's exclusion zone would block the site's only emergency vehicle route?

Answer

The plan must identify an alternate emergency access route or a procedure to quickly open the exclusion zone if needed. Treating the exclusion zone as an absolute priority ignores the risk of delayed emergency response elsewhere. Shrinking the zone below the crane's real swing radius is not an acceptable fix.

#291 · Lift Planning and Site Setup

Why is documenting the actual mat size used on lift day important?

Answer

Without this documentation, there is no way to confirm the calculated ground bearing pressure was actually achieved on site. Using a smaller mat than planned produces higher real pressure than calculated. This verification gap is a genuine safety concern, not just paperwork.

#292 · Lift Planning and Site Setup

What extra step might be needed on a site with a documented history of utility strikes?

Answer

The crew should consider hand digging test pits or other extra verification methods beyond standard electronic locates. Standard maps and locates are not always fully accurate. This extra step targets a known high-risk area specifically.

#293 · Lift Planning and Site Setup

What should a backup communication plan include?

Answer

It should include a pre-agreed backup method, such as standardized hand signals, and a clearly defined action, like an immediate stop, if communication is lost entirely. Assuming radios rarely fail ignores a real and foreseeable risk. Shouting is not a standardized substitute.

#294 · Lift Planning and Site Setup

What should a signal person do if they lose sight of the load unexpectedly?

Answer

They should signal an immediate stop until a clear line of sight is restored or another spotter with a clear view takes over. Continuing to signal based on the load's last known position risks directing the operator into an unseen hazard. Waiting silently without calling a stop leaves the load moving uncontrolled.

#295 · Lift Planning and Site Setup

How should a crew choose between two candidate crane setup positions with different tradeoffs?

Answer

Evaluate both ground bearing pressure relative to soil rating and rated capacity at radius for each position. Choose the option where both factors are satisfied with adequate margin. A single-factor decision, like always choosing shorter radius, ignores the other requirement.

#296 · Lift Planning and Site Setup

What must occur if a planned line de-energization or line cover has not been completed by lift day?

Answer

The lift must be delayed until the utility confirms the de-energization or line cover is actually in place. Falling back to the standard minimum approach distance changes the risk basis the plan was built around. Crews should never attempt to install line covers themselves.

#297 · Lift Planning and Site Setup

What core topics should never be skipped in a pre-lift briefing, even under time pressure?

Answer

The communication method, the signal person's identity, and the emergency stop procedure must always be covered. These are the core elements keeping the crew safely coordinated. Unrelated details, like crane manufacturing history, can be shortened instead.

#298 · Lift Planning and Site Setup

Why does a tandem lift require only one active signal source?

Answer

Two active signal sources on a tandem lift risk the two cranes receiving conflicting instructions at the same time. Only the designated lead signal person should give active signals to both operators. If a second source starts signaling, the lift should stop until this is corrected.

#299 · Lift Planning and Site Setup

Why is using total crane weight instead of individual outrigger reaction loads a flawed ground bearing pressure method?

Answer

Individual outrigger reaction loads vary based on load position and crane configuration during a specific lift. Dividing total weight evenly can significantly understate pressure at the most heavily loaded outrigger. Reaction loads are not identical across all outriggers in most configurations.

#300 · Lift Planning and Site Setup

What should a crew do if a nearby line's voltage cannot be confirmed and the utility is unreachable before the lift?

Answer

Delay the lift until the voltage is confirmed, or apply the largest minimum approach distance on the clearance table until it is. Using the lowest voltage distance to save time risks an unsafe clearance. An unreachable utility does not remove the underlying safety requirement.

#301 · Lift Planning and Site Setup

What is the risk of an out-of-level crane that develops mid-lift due to outrigger settlement?

Answer

Continued settlement can push the crane further out of level as the lift progresses, increasing side-load risk. The crew should stop at the earliest safe point, lower the load if possible, and re-level before continuing. A pre-lift check does not remain valid once conditions physically change.

#302 · Lift Planning and Site Setup

What should a crew do if setup discovers a second, previously unmarked power line?

Answer

Halt setup, identify the second line's voltage and required clearance, and revise the full lift plan to address both lines. Proceeding under the original single-line plan leaves the second hazard unassessed. Guessing that it matches the first line's voltage is not reliable.

#303 · Lift Planning and Site Setup

Why can rain during a multi-hour lift sequence create new risk even after ground was already checked?

Answer

Rain can reduce the soil's actual bearing capacity below the originally calculated safe margin as it saturates. The crew should reassess ground conditions as weather changes during the lift. Pausing or adding matting may be needed partway through.

#304 · Lift Planning and Site Setup

How should a crew resolve differing jurisdictional minimum approach distance standards for the same voltage line?

Answer

Confirm which standard actually applies to the current site and apply that standard's required distance. Choosing whichever distance is easier to achieve prioritizes convenience over the actual legal requirement. Averaging two different standards' figures is not a valid method.

#305 · Lift Planning and Site Setup

Why must a crew avoid informal hand signal shortcuts developed between one operator and one signal person?

Answer

Other crew members, spotters, or inspectors present at the site must also be able to correctly interpret every signal given. A strong safety record between two individuals does not guarantee others understand the shortcuts. Standard charts protect everyone on site, not just the established pair.

#306 · Lift Planning and Site Setup

What should a crew consider before shrinking an exclusion zone to match a shorter planned working radius?

Answer

Whether the crane could still reach a longer radius at any point during the lift sequence, such as while repositioning the load. A barricade sized only for the shortest radius could leave a gap if the boom moves further out. This should be reviewed, not assumed.

#307 · Lift Planning and Site Setup

What is the correct response if a load begins swaying unexpectedly near a power line boundary?

Answer

Stop all boom and load movement immediately and allow the sway to settle before adjusting position. Continuing to lower or hoist quickly does not address the uncontrolled sway. Resume only once control is regained and clearance is reconfirmed.

#308 · Lift Planning and Site Setup

How should a conflict between ground bearing pressure limits and power line clearance requirements be resolved?

Answer

Engineer a solution, such as improved matting, that satisfies both requirements at once, or reconsider the entire lift approach if no position works for both. Both requirements exist to prevent different, serious hazards. Satisfying only one while ignoring the other leaves a real risk.

#309 · Lift Planning and Site Setup

How should a crew weigh current weather conditions against a worsening forecast before starting a lift?

Answer

Assess whether the lift can realistically be completed or safely paused before forecasted conditions deteriorate. Relying only on current acceptable conditions ignores a foreseeable, specific hazard already known in advance. Delay starting if safe completion cannot be reasonably assured.

#310 · Lift Planning and Site Setup

What is the best long-term fix for radio interference from another crane operating nearby?

Answer

Coordinate with the other crew to assign separate, non-overlapping channels, then verify the fix with a fresh radio check. Simply repeating instructions more often does not resolve the underlying overlap. Increasing transmission power does not fix a shared-frequency problem.

#311 · Lift Planning and Site Setup

What should be done after a near-miss where a load briefly entered the minimum approach distance zone?

Answer

The near-miss should be formally investigated to find what allowed the boundary approach, and corrective action taken before the next similar lift. Concluding no action is needed because no contact occurred ignores that the same cause could produce contact next time. It should always be documented.

#312 · Lift Planning and Site Setup

What must a crew do if only a different crane model with a similar load chart is available on lift day?

Answer

Obtain and apply the actual load chart for the specific model present, and revise the plan if its capacity differs. Using the original plan's figures assumes an equivalence that may not hold. Physical ability to reach the load does not confirm rated capacity is sufficient.

#313 · Lift Planning and Site Setup

What planning approach fits a lift where several safety factors are each satisfied but with very little margin?

Answer

Build in active, continuous monitoring of all factors throughout the lift, with clearly defined stop triggers for each. A small change in any one factor could push the whole lift outside safe limits. A single pre-lift check is not enough when margins are this thin.

#314 · Lift Planning and Site Setup

How should a crew resolve a signal person having good load visibility but poor exclusion zone visibility?

Answer

Assign an additional dedicated perimeter watch person so both the load and the boundary are actively monitored. Accepting the tradeoff leaves the pedestrian entry risk unaddressed. Moving the signal person to the perimeter instead just shifts the same gap onto the load.

#315 · Lift Planning and Site Setup

Why can't a site survey completed a week earlier always be trusted on lift day itself?

Answer

Ground conditions can change significantly due to weather or other factors in the time since the survey. The crew must reassess current conditions on lift day, especially if visible standing water or soft ground appears. Trusting an old survey over visible current evidence is not safe.

#316 · Lift Planning and Site Setup

What lesson should be drawn if a lift plan sign-off was skipped but the lift went well anyway?

Answer

The sign-off step must still be enforced going forward, since its value is catching errors before the lift, not confirming after the fact that things went well. A single successful outcome does not prove the control is unnecessary. Treating sign-off as optional for experienced crews is not a safe assumption.

#317 · Lift Planning and Site Setup

What role does a spotter play near an energized line, and why shouldn't they take on other tasks?

Answer

A dedicated spotter's sole task is watching the required minimum approach distance to the line throughout the lift. Assigning them a second task, like guiding the load, divides their attention during a critical moment. A separate person should be assigned to guide the load instead.

#318 · Lift Planning and Site Setup

Why does a crew need to check for buried utilities even when crossing a site roadway?

Answer

Roadways commonly have buried gas, electrical, or communication lines running beneath or alongside them. Utility locates must be obtained before the crane travels across any ground, including roadways. Daylight travel alone does not reveal underground hazards.

#319 · Lift Planning and Site Setup

What must a lift plan for a rooftop landing address beyond the ground-level swing radius?

Answer

It must address the rooftop landing zone itself and any ground area beneath the load's travel path where a dropped load could fall. Limiting the exclusion zone to only ground-level swing ignores the vertical hazard path leading up to the rooftop. Both areas need separate hazard control.

#320 · Lift Planning and Site Setup

How should multi-contractor site coordination handle an upcoming crane lift?

Answer

Other trades must be notified of the lift's schedule and exclusion zone boundaries so their workers and equipment stay clear. Limiting coordination to only the crane crew's internal schedule ignores the risk of other trades wandering into the hazard area. A posted schedule and direct supervisor contact are common methods.

#321 · Crane Assembly Disassembly and Transport

What is a pinch point?

Answer

A pinch point is any spot where a moving part can trap a worker against a fixed object or another moving part. Boom pin bores closing together during assembly are a classic example. Keep hands and fingers clear and use tools to guide alignment.

#322 · Crane Assembly Disassembly and Transport

Why does a mobile crane need counterweight?

Answer

Counterweight creates an opposing moment that balances the tipping force of a suspended load out over the front of the crane. Without enough counterweight, the crane can tip before it reaches its rated capacity. It must be installed and locked before lifting.

#323 · Crane Assembly Disassembly and Transport

In what order should boom section pins be removed during disassembly?

Answer

Pins are removed from the tip section inward, one section at a time, only after each freed section is properly cribbed or supported. Removing base pins first while tip sections hang unsupported is unsafe. Order preserves the boom's load path.

#324 · Crane Assembly Disassembly and Transport

What must jib pins meet before a jib goes into service?

Answer

Jib pins must be the exact rating, size, and type specified by the crane manufacturer for that connection. Substituting a lighter, unrated, or improvised pin can cause the boom-point connection to fail under load. Bright paint helps visibility but is not a substitute for correct rating.

#325 · Crane Assembly Disassembly and Transport

What is the main job of an escort vehicle on an oversize load move?

Answer

An escort vehicle warns approaching traffic and helps the crane carrier maintain safe clearance and lane position through tight spots. It is not responsible for mechanical inspections or fuel management. Escort duties are set by the transport permit.

#326 · Crane Assembly Disassembly and Transport

When does a load count as oversize for highway transport?

Answer

A load is oversize when it exceeds the legal width, height, or length limits set by the jurisdiction it is travelling through. Weight relative to the tow vehicle or trip distance does not define oversize status. Oversize loads require permits and often escorts.

#327 · Crane Assembly Disassembly and Transport

Why are axle load limits enforced for crane carriers and trailers?

Answer

Axle load limits protect road surfaces and bridge structures from excessive point loading that they were not engineered to handle. They are unrelated to sightlines or crane warranty terms. Weigh stations check compliance along the route.

#328 · Crane Assembly Disassembly and Transport

What must be done with outriggers before a crane travels on a public road?

Answer

Outriggers must be fully retracted and locked in their stowed travel position before the crane moves. Extended beams exceed legal travel width and create a serious collision hazard. This applies regardless of trip distance.

#329 · Crane Assembly Disassembly and Transport

What does travel position mean for a crane's boom?

Answer

Travel position is the specific boom angle and length the manufacturer designates as safe for road movement. It controls the crane's centre of gravity and reduces wind exposure during transport. It is not left to operator preference.

#330 · Crane Assembly Disassembly and Transport

Why is a tag line attached to a boom section being lifted into place?

Answer

A tag line lets ground crew apply controlled tension to guide the section and prevent uncontrolled rotation or swing. It has no electronic function and does not replace spotters watching the whole lift. It cannot measure weight.

#331 · Crane Assembly Disassembly and Transport

What is blocking or cribbing used for during boom disassembly?

Answer

Blocking supports a boom section set temporarily on the ground so it cannot shift, roll, or sink unevenly. It is purely temporary support, not a strength upgrade or grounding device. It keeps the section stable until the next step.

#332 · Crane Assembly Disassembly and Transport

What does self-erection mean for a mobile crane?

Answer

Self-erection means the crane raises its own boom sections using its own boom hoist and winch system, without a separate assist crane. An operator and hydraulic power are still required throughout. Ground bearing capacity must support the added loads generated.

#333 · Crane Assembly Disassembly and Transport

What does a transport permit authorize?

Answer

A transport permit authorizes a specific oversize or overweight move and sets binding conditions such as approved route, travel window, and required escorts. Deviating from those conditions can void the permit and create legal liability. It does not replace mechanical inspection.

#334 · Crane Assembly Disassembly and Transport

What is pick-and-carry, and how does it differ from assembly or transport?

Answer

Pick-and-carry is moving a suspended load short distances across a work site while the crane travels. It is a distinct on-site lifting operation, not related to boom assembly or highway transport. Confusing the terms can lead to the wrong safety procedure being applied.

#335 · Crane Assembly Disassembly and Transport

Why must the boom be at a low travel angle before highway travel?

Answer

A low, manufacturer-specified travel angle reduces wind loading on the boom and keeps the crane's centre of gravity lower and more stable. This requirement applies regardless of counterweight status. It is unrelated to outrigger position.

#336 · Crane Assembly Disassembly and Transport

What does the counterweight locking pin do?

Answer

The counterweight locking pin physically restrains stacked slabs so they cannot shift, slide, or fall during operation or travel. It has no role in engine speed or hydraulic cooling. Without it, slabs are not considered securely mounted.

#337 · Crane Assembly Disassembly and Transport

What is the purpose of chains, binders, and straps on a transport trailer?

Answer

They are load securement devices that keep crane components from shifting or coming loose while the trailer moves on the highway. They do not change the trailer's registered gross weight. Both boom sections and counterweight require proper securement.

#338 · Crane Assembly Disassembly and Transport

What protects workers from falls during boom assembly at height?

Answer

Approved fall protection equipment, such as engineered anchor points and a full body harness, is the primary control. Working faster or a verbal reminder does not eliminate the fall hazard. Positioning someone to catch a falling worker is not a real control.

#339 · Crane Assembly Disassembly and Transport

What must be confirmed before a boom pin is driven home?

Answer

Both pin bores must be fully aligned and everyone's hands must be clear of the closing pinch point. Releasing crane tension too early risks the sections shifting out of alignment. This confirmation happens before, not after, driving the pin.

#340 · Crane Assembly Disassembly and Transport

What does a locking pin retainer prevent?

Answer

A retainer prevents the boom pin from vibrating loose or backing out of the bore during operation. It has no effect on section weight and does not replace correct torque procedures. Losing a retainer can allow a structural pin to walk out.

#341 · Crane Assembly Disassembly and Transport

Why must counterweight slabs follow the manufacturer's installation sequence?

Answer

The specified order and orientation are engineered for the crane's stability and locking geometry. Installing the heaviest slab last to lock others ignores that engineering. Skipping locking pins at any stage is never acceptable.

#342 · Crane Assembly Disassembly and Transport

Are stacked counterweight slabs secure before retaining pins are installed?

Answer

No. Slabs have no mechanical restraint until the retaining pins engage, even if they appear to sit securely. This is a physical stability issue, not related to axle rating. Pins must be installed before slabs count as part of a stable configuration.

#343 · Crane Assembly Disassembly and Transport

What must be verified before raising a jib?

Answer

The load chart must permit the intended jib length and offset angle for that configuration, since jib geometry changes the crane's rated capacity. Boom length and counterweight status do not confirm this on their own. Outrigger position is unrelated to this specific check.

#344 · Crane Assembly Disassembly and Transport

What is the function of jib pendant lines?

Answer

Pendant lines connect the jib strut to the jib and control its angle relative to the main boom as it is raised. They carry no hydraulic power and are not part of any cooling system. They are not the main hoist line.

#345 · Crane Assembly Disassembly and Transport

Why must the boom stop swinging if jib backstop pins are not yet installed?

Answer

Without backstop pins, the jib lacks its intended fall-back restraint, so swinging must stop until the pins are properly installed and secured. Jib length under ten metres does not remove this requirement. Installing pins after swinging defeats the backstop's purpose.

#346 · Crane Assembly Disassembly and Transport

What must be confirmed before a freed boom section is lowered?

Answer

The section must be properly rigged and supported by the assist crane or hoist line before lowering begins. Painted identification numbers and crew standoff distance are unrelated safety controls here. Shutting off the carrier engine is not required for this step.

#347 · Crane Assembly Disassembly and Transport

What hazard does a tag line control while lowering a boom section?

Answer

It controls the section so it does not swing or rotate unpredictably and strike nearby workers as it descends. Hydraulic overheating and tire pressure are unrelated hazards. This is specific to suspended, unguided loads.

#348 · Crane Assembly Disassembly and Transport

What must be confirmed about ground conditions before self-erection begins?

Answer

The ground must be able to bear the outrigger and crawler loads generated as the crane raises its own boom. Ground colour and drainage slope are not relevant to bearing capacity. Sweeping away loose gravel alone does not confirm adequate support.

#349 · Crane Assembly Disassembly and Transport

What do erection cables do during self-erection rigging?

Answer

Erection cables, reeved from the winch drum over a gantry or A-frame to a point along the boom, supply the lifting force to raise the boom without a separate assist crane. They serve no grounding function and do not replace boom pins. Boom pins still secure sections structurally.

#350 · Crane Assembly Disassembly and Transport

What must be verified before lifting operations begin after self-erection?

Answer

All erection cables and rigging used only for assembly must be removed, since leftover rigging can interfere with the boom or hoist line. Counterweight should be installed per the load chart, not left off. Outriggers must be set for lifting, not left stowed.

#351 · Crane Assembly Disassembly and Transport

What document is required before moving an oversize crane load on public roads?

Answer

An oversize load transport permit issued by the relevant transport authority is required before the move proceeds. A manufacturer waiver, updated load chart, or counterweight inspection certificate does not satisfy this legal requirement. The permit sets the move's binding conditions.

#352 · Crane Assembly Disassembly and Transport

Why must permit travel windows and approved routes be followed exactly?

Answer

Deviating from them can void the permit entirely and expose the carrier to legal liability if an incident occurs. These conditions apply regardless of escort availability. They are legal obligations, not optional suggestions.

#353 · Crane Assembly Disassembly and Transport

What typically happens if a carrier exceeds axle load limits at a weigh station?

Answer

The load is usually delayed, fined, or required to redistribute weight before continuing the trip. There is no automatic permit extension or shifting of excess weight to the escort vehicle. Compliance is checked and enforced right at the scale.

#354 · Crane Assembly Disassembly and Transport

What is the role of a trailing escort vehicle?

Answer

A trailing escort monitors rear clearance and warns following traffic that a wide load is ahead, helping prevent rear-end collisions or unsafe passing. It is not assigned to carry fuel or perform inspections. It mirrors the lead escort's role at the front.

#355 · Crane Assembly Disassembly and Transport

What does a boom cradle support do during transport?

Answer

It carries the boom's weight and reduces stress from road vibration over the length of the trip. It has no effect on lifting capacity and does not replace the requirement to retract outriggers separately. The boom must be fully seated, not partially extended.

#356 · Crane Assembly Disassembly and Transport

How should a disassembled boom section be secured on a flatbed trailer?

Answer

Rated chains or straps must be placed at multiple points, following the applicable load securement standard. A single chain at the midpoint or friction alone cannot reliably restrain a heavy, rigid section against road forces. Proper securement prevents shifting or falling in transit.

#357 · Crane Assembly Disassembly and Transport

Why is load height checked against route clearances before a trip?

Answer

The check avoids striking overhead structures such as bridges and power lines. Brake light function, fuel level, and tire pressure are separate pre-trip items. A height violation can cause a catastrophic collision with fixed infrastructure.

#358 · Crane Assembly Disassembly and Transport

What does engaging the swing lock accomplish before road travel?

Answer

It prevents the crane's upperworks from rotating unexpectedly while the vehicle moves, which could shift the centre of gravity or strike nearby objects. It has no effect on legal travel speed. It is a standard step alongside outrigger stowage.

#359 · Crane Assembly Disassembly and Transport

Why must boom rest pads be properly seated before travel?

Answer

Properly seated pads reduce vibration damage to the boom and keep it from shifting off the cradle during highway travel. This step is separate from anti-two-block calibration or boom hoist tensioning. Outriggers must be fully retracted, not partially extended, for travel.

#360 · Crane Assembly Disassembly and Transport

What must be confirmed about an assist crane's footing before it takes a final boom section's weight?

Answer

Its outrigger or crawler footing must be confirmed stable enough to support the additional load safely. Fuel level, painted stickers, and wind speed above ten kilometres per hour are not the relevant checks here. Ground support directly affects tipping and sinking risk.

#361 · Crane Assembly Disassembly and Transport

Why is a dedicated signal person used during boom assembly?

Answer

A signal person gives the assist crane operator clear, consistent direction, which matters most when the operator's sightlines are limited. This role does not reduce crew size or replace tag lines. Consistent signals prevent conflicting instructions during unsafe moves.

#362 · Crane Assembly Disassembly and Transport

Is it ever acceptable to walk under a suspended boom section?

Answer

No. Workers must wait until the section is set down or properly blocked before entering that space. Wearing protective gear does not make walking under a suspended load acceptable. Partial pinning at one end does not make it safe either.

#363 · Crane Assembly Disassembly and Transport

What should a crew do if boom pin bores are misaligned?

Answer

Adjust the assist crane's positioning so the sections line up properly before attempting the pin again. Forcing a pin with a heavier hammer, leaving it partially seated, or substituting an undersized pin can all damage or weaken the connection. Alignment must be correct before driving the pin.

#364 · Crane Assembly Disassembly and Transport

In what order should counterweight slabs be removed?

Answer

Generally in reverse order from how they were installed, following the manufacturer's chart, to maintain crane stability throughout. Removing them in any order or all pins at once ignores the engineered sequence. Reverse order preserves the same stability logic used during installation.

#365 · Crane Assembly Disassembly and Transport

What must the rigging do throughout jib lowering?

Answer

The boom hoist and jib rigging must maintain controlled tension so the jib descends smoothly and predictably. Releasing tension or disconnecting rigging partway through removes control at the most hazardous point. Tension control matters regardless of jib length.

#366 · Crane Assembly Disassembly and Transport

Why is the hook block removed and stored separately for transport?

Answer

An attached block can swing freely and strike objects or workers during road travel. This has nothing to do with fuel efficiency or boom hoist calibration. It applies to every transport move, not only tall loads.

#367 · Crane Assembly Disassembly and Transport

What is the purpose of a pre-departure walk-around inspection?

Answer

It catches loose pins, unsecured components, or damage before the crane travels on the highway, where such issues become far more dangerous. It does not substitute for load chart calculations or certification checks. It focuses on the physical condition of the crane and attachments.

#368 · Crane Assembly Disassembly and Transport

Why is cribbing used under a temporarily set-down boom section?

Answer

Cribbing prevents the section from sinking, tipping, or shifting on soft or uneven ground during assembly pauses. It has no bearing on future reuse of the section or when retainers get installed. Its purpose is purely to keep the section stable in place.

#369 · Crane Assembly Disassembly and Transport

Why should removed boom pins be stored in a labeled container?

Answer

It prevents pins from being lost, mixed up between sections, or left underfoot as a trip hazard. It does not reduce the number of pins needed or allow cross-model reuse. Organized storage protects both safety and reassembly accuracy.

#370 · Crane Assembly Disassembly and Transport

What does a jib backstop system do?

Answer

It limits how far the jib can fall back if the boom is lowered sharply, protecting the structure and nearby workers. It does not increase the jib's rated capacity or replace the pendant lines. Backstops and pendants work together, not as substitutes.

#371 · Crane Assembly Disassembly and Transport

Why does uneven weight distribution across trailer axles matter?

Answer

It can overload individual axles beyond their rated limit, affecting handling, braking, and legal compliance. This is a real structural and legal concern, not a cosmetic issue. Tarping the load does not fix weight distribution.

#372 · Crane Assembly Disassembly and Transport

What must a driver confirm before departure regarding overpass clearance?

Answer

The load's measured travel height must clear every overpass on the planned route, confirmed before the trip begins. This cannot be verified once the load is already underway. Missing this check risks a direct structural collision.

#373 · Crane Assembly Disassembly and Transport

Why is standing between two boom sections during final approach dangerous?

Answer

It exposes a worker to being crushed if the sections come together faster or differently than expected. This position offers no sightline advantage that outweighs the crush risk. Workers should direct the lift from a position clear of the closing gap.

#374 · Crane Assembly Disassembly and Transport

Why use a spreader bar when lifting counterweight slabs?

Answer

A spreader bar keeps lifting slings at a safer angle and helps control the slab's orientation as it is set into place. It does not eliminate the need for a signal person or reduce the slab's actual weight. Retaining pins are still required after the lift.

#375 · Crane Assembly Disassembly and Transport

Why must outriggers be both fully retracted and fully raised before travel?

Answer

Outriggers that are not fully stowed in both directions can exceed legal travel width or strike obstacles along the route. This is a direct road safety and legal compliance issue. No load chart provision allows partially stowed outriggers for highway travel.

#376 · Crane Assembly Disassembly and Transport

Why assign a clear signal-directing role before assembly begins?

Answer

It avoids confusion and conflicting signals during a lifting task where miscommunication can cause a serious incident. It does not reduce the number of pins needed or excuse the crew from a pre-task safety meeting. Clear roles are core planning for high-hazard lifts.

#377 · Crane Assembly Disassembly and Transport

What does a load securement inspection confirm?

Answer

It confirms that devices such as chains, binders, and straps actually restrain the load from shifting, falling, or loosening in transit. Operator certification and load chart matching are checked separately. A functioning escort radio is unrelated to physical securement.

#378 · Crane Assembly Disassembly and Transport

Why is boom cross bracing removed incrementally during disassembly?

Answer

Removing bracing one section at a time maintains the boom's structural stability throughout teardown. This approach is not primarily about saving time. It does not eliminate the need for cribbing or a signal person.

#379 · Crane Assembly Disassembly and Transport

What must be confirmed before the first test lift after assembly?

Answer

That all assembly rigging has been removed, so nothing interferes with the boom or hoist line during the lift. This does not change the load chart rating or allow skipping counterweight installation. Leftover rigging is a common and preventable hazard.

#380 · Crane Assembly Disassembly and Transport

What does a weigh scale check confirm for a counterweight trailer?

Answer

It verifies the axle and gross weights comply with the limits set in the transport permit. It is not used to confirm paint markings or calculate fuel consumption. Weight compliance is a core, enforceable condition of the permit.

#381 · Crane Assembly Disassembly and Transport

Why must pin retaining bolts be torqued to the manufacturer's specification?

Answer

Correct torque keeps the retainer secure without over-stressing the pin or its bracket, which could cause premature failure. It does not change a section's rated lifting capacity or eliminate the need for periodic inspection. This applies to boom pins generally, not only counterweight pins.

#382 · Crane Assembly Disassembly and Transport

What should a carrier driver do when an escort warns of a narrow bridge ahead?

Answer

Slow down and follow the escort's guidance to position the load safely through the constriction. Ignoring the message or speeding up increases risk. The escort exists specifically to manage exactly this kind of hazard.

#383 · Crane Assembly Disassembly and Transport

Why load counterweight onto a dedicated trailer separate from the crane carrier?

Answer

It helps keep both the carrier's and the trailer's individual axle and gross weights within legal transport limits. It does not reduce required permits or eliminate the need for load securement or escorts. Weight distribution across vehicles is a common compliance strategy.

#384 · Crane Assembly Disassembly and Transport

What ground bearing check is required after overnight rain before self-erection continues?

Answer

Softened ground means the crew must reassess bearing capacity before the next lift, even if outriggers were set safely the day before. A successful earlier lift does not confirm current ground conditions after weather changes. Both main and assist crane footing must be evaluated.

#385 · Crane Assembly Disassembly and Transport

What is the manufacturer-intended order for connecting jib pendants and pinning the jib?

Answer

The jib's heel pin is typically installed first to create the structural pivot, pendants are then attached to control the jib's angle as it is raised, and a separate offset-angle lock pin is engaged once the jib is positioned. Pinning before pendants are attached leaves the jib without control during raising. The jib strut alone does not manage the angle without the pendant system.

#386 · Crane Operation

What does LMI stand for?

Answer

LMI stands for Load Moment Indicator. It is a system that calculates the crane's actual loading as a percentage of its rated capacity at the current boom angle, length, and radius. It gives the operator a continuous readout so overloading can be caught before it happens.

#387 · Crane Operation

What is an anti-two-block device?

Answer

An anti-two-block device is a safety switch that stops hoist and boom functions before the hook block or ball contacts the boom tip sheave. Contact at that point, called two-blocking, can snap the wire rope or cause a sudden load drop. The device usually uses a weighted trip lever near the boom tip.

#388 · Crane Operation

The LMI alarm sounds continuously at 100% of rated capacity during a lift. What should the operator do?

Answer

The operator should stop all crane motion immediately and not raise the load, extend the boom, or increase the radius. The load should be lowered or the radius decreased under control to bring the reading back under the limit. Continuing the lift risks structural failure or tipping.

#389 · Crane Operation

What is the purpose of a load chart?

Answer

A load chart shows the maximum weight a crane can safely lift at a given boom length, boom angle, and radius, along with the crane's configuration such as outrigger spread and counterweight. The operator must read the chart before every lift to confirm the load is within capacity. Load charts differ by crane model and setup, so the correct chart for that exact configuration must be used.

#390 · Crane Operation

What is radius in the context of a load chart?

Answer

Radius is the horizontal distance measured from the crane's centre of rotation to the centre of the suspended load or hook. As radius increases, the crane's lifting capacity decreases because the load creates more tipping moment. Radius changes when the boom angle lowers or the boom telescopes out.

#391 · Crane Operation

Why does lifting capacity drop as boom radius increases?

Answer

A longer radius creates a longer lever arm between the load and the crane's centre of rotation, which increases the tipping moment on the crane. The crane's structure and counterweight can only resist a limited amount of moment before it becomes unstable. That is why load charts show lower capacities at greater radii even though the boom and rigging have not changed.

#392 · Crane Operation

What is a rated capacity limiter (RCL) and how does it differ from a simple indicator?

Answer

A rated capacity limiter is a system that not only displays the load moment percentage but automatically cuts out or stops crane functions that would increase loading once the rated capacity is reached. A simple load moment indicator only warns the operator with an alarm and leaves the response up to them. Many modern cranes combine both functions in one system.

#393 · Crane Operation

What is the standard one-arm Stop signal, and who can give it?

Answer

The Stop signal is given by extending one arm out to the side, palm down, and moving it back and forth horizontally. It tells the operator to stop the motion currently in progress. Only the designated signal person, or a qualified substitute, should give routine Stop signals during normal operation.

#394 · Crane Operation

Who is allowed to give signals to the crane operator during a lift?

Answer

Only the designated signal person may give working signals such as hoist, lower, swing, and boom movement, unless the load is out of the operator's direct view in which case a signal person is mandatory. Any person on site can give an emergency stop signal at any time. The operator should not act on signals from anyone other than the assigned signal person for normal lift movements.

#395 · Crane Operation

The operator loses visual contact with the signal person mid-lift. What is the correct response?

Answer

The operator should stop all crane motion immediately and hold the load in place until visual or radio contact with the signal person is restored. Continuing to move the load blind risks striking workers, equipment, or structures. Once contact is regained, the operator can resume following confirmed signals.

#396 · Crane Operation

What is the advantage of radio communication over hand signals for a crane lift?

Answer

Radio communication allows clear verbal instructions when the signal person is out of the operator's line of sight or too far away for hand signals to be seen clearly. It also allows more detailed direction, such as exact stopping points or hazard warnings. Standard hand signals are still required as a backup in case radios fail.

#397 · Crane Operation

What should be confirmed before starting any radio-signalled lift?

Answer

The operator and signal person should confirm they are on the same channel, that the radios are working clearly, and agree on a backup plan such as hand signals or an immediate stop if radio contact is lost. A quick radio check before the lift begins prevents miscommunication once the load is in motion. This confirmation should happen at the start of every shift and after any radio or personnel change.

#398 · Crane Operation

What does coordinating hoist and boom functions together allow the operator to do?

Answer

Coordinating hoist and boom together lets the operator move a load in a controlled path, such as lifting while swinging, without the load swinging out unpredictably or dragging along the ground. Smooth simultaneous control keeps the load steady and reduces side loading on the boom. Jerky or single-function-at-a-time movement on a coordinated lift often causes the load to swing.

#399 · Crane Operation

What is boom telescoping and when is it typically used during a lift?

Answer

Boom telescoping is extending or retracting the sections of the boom to change its length and reach. It changes both the radius and the capacity available from the load chart, so it must be done carefully and rechecked against the chart. Telescoping under load is riskier than adjusting boom angle and should only be done within the limits the manufacturer allows.

#400 · Crane Operation

Why is swinging a load smoothly rather than abruptly important?

Answer

Abrupt swing starts or stops cause the load to swing out on the hook like a pendulum, which changes the actual radius and can exceed the rated capacity or strike nearby objects. Smooth acceleration and deceleration keep the load hanging plumb under the hook. This is especially important with long wire rope falls where swing amplifies quickly.

#401 · Crane Operation

What is a tandem lift?

Answer

A tandem lift, also called a multi-crane lift, uses two or more cranes to lift a single load together because it is too long, heavy, or awkward for one crane alone. It requires a detailed lift plan showing each crane's share of the load and its own load chart check. Tandem lifts carry higher risk because the cranes must move in coordination.

#402 · Crane Operation

Why must each crane's share of the load be calculated separately in a tandem lift?

Answer

The load is rarely split evenly between the cranes because the centre of gravity is usually not exactly between the two lift points. Each crane must be checked against its own load chart using its actual calculated share, including a safety margin for uneven weight shifts during the lift. Underestimating one crane's share can overload it even if the total load seems within combined capacity.

#403 · Crane Operation

What is the role of a single, clearly identified signal person in a tandem lift?

Answer

In a tandem lift one signal person directs both crane operators so their movements stay synchronized and neither crane moves independently. Using more than one uncoordinated signal person risks conflicting instructions that could cause the load to tip or one crane to take on excess weight. All operators involved must agree in advance on who that signal person is.

#404 · Crane Operation

What extra planning step is required before any tandem lift that is not needed for a single-crane lift?

Answer

A written lift plan is required showing rigging points, each crane's load share, boom positions, ground conditions, and communication procedures. The plan should be reviewed with both crews before the lift begins. This extra planning accounts for the added complexity of keeping two machines synchronized around one load.

#405 · Crane Operation

What should an operator do if wind speeds approach the manufacturer's rated limit during a lift?

Answer

The operator should stop lifting operations and secure the load and boom before wind reaches the rated limit, not after. Wind increases the effective load on a crane by pushing on both the boom and the suspended load, especially large flat loads that act like a sail. Manufacturer wind limits are found in the crane's operating manual and must never be exceeded.

#406 · Crane Operation

Why are large flat loads, like sheet material, especially dangerous to lift in windy conditions?

Answer

A large flat surface catches wind like a sail, which adds significant unpredictable side force to the load. This side force can swing the load into structures or workers and can also add stress the crane's load chart does not account for. Tag lines and reduced wind thresholds are often required for these loads.

#407 · Crane Operation

What is the purpose of tag lines during outdoor lifts?

Answer

Tag lines are ropes attached to the load that ground workers use to control its rotation and swing, especially in wind. They let workers guide the load without touching it directly, keeping hands clear of pinch points. Tag lines should be long enough to keep the handler outside the load's swing radius.

#408 · Crane Operation

What should an operator check about ground and weather conditions before operating in rain or after a storm?

Answer

The operator should check that outrigger footing is stable and has not softened from rain, that visibility is adequate, and that lightning is not in the area. Wet or saturated ground can shift or settle under outrigger pads during a lift, causing the crane to tip. Operations should stop if lightning is within the site's established safe distance.

#409 · Crane Operation

What is the correct response if lightning is detected near an outdoor lift site?

Answer

All crane operations should stop immediately and the operator should move to a safe location away from the crane structure. Lightning can travel down the boom and through the crane to the ground, putting the operator and ground crew at serious risk. Work should not resume until the site's lightning safety policy allows it, typically after a set time with no further strikes.

#410 · Crane Operation

What does an overload alarm indicate and what is the immediate operator response?

Answer

An overload alarm means the crane has reached or exceeded its rated capacity for the current configuration. The operator must stop increasing load, lower the load or reduce radius smoothly, and never override the alarm to continue lifting. Ignoring the alarm risks structural damage or a tip-over.

#411 · Crane Operation

What should an operator do if hydraulic pressure loss occurs while a load is suspended?

Answer

The operator should stop all motion immediately and engage any available holding brakes, avoiding any action that relies on the lost hydraulic pressure until the cause is identified. Lowering the load slowly is acceptable only if a manufacturer-specified emergency or manual lowering system allows controlled lowering without relying on the lost pressure source. If safe lowering is not possible, the area beneath and around the load must be cleared and barricaded until repairs are made. Trying to force a function during a pressure loss risks sudden uncontrolled movement.

#412 · Crane Operation

What is the correct action if the crane experiences a total power failure with a load in the air?

Answer

The operator should apply the hoist brake and hold position, avoid attempting further movement, and notify others on site immediately. Most cranes use spring-applied hydraulically-released brakes that hold the load automatically when power is lost. The area under the load should be cleared and kept clear until power is restored and the load can be safely set down or lowered.

#413 · Crane Operation

Why do most crane hoist brakes fail in the 'on' or applied position rather than releasing?

Answer

Hoist brakes are designed as spring-applied, power-released systems so that a loss of hydraulic or electrical power causes the brake to engage automatically rather than release. This fail-safe design prevents a suspended load from free-falling if power is lost. It is a deliberate engineering safeguard rather than a coincidence.

#414 · Crane Operation

What should an operator do before leaving a crane unattended, even briefly, with no load on the hook?

Answer

The operator should lower the boom to a safe position or rest it on the boom rest if equipped, apply all parking brakes and swing lock, and remove the key or disable the controls. The hook should be raised close to the boom tip and secured so it cannot swing freely. Leaving a crane unattended without securing it creates a hazard for anyone nearby.

#415 · Crane Operation

What is the correct end-of-shift shutdown procedure for a mobile crane?

Answer

The operator should lower the boom, retract it if applicable, set the swing lock, lower outriggers only after the boom is secured, and shut down the engine following the manufacturer's sequence. All controls should be returned to neutral and the crane should be left in a stable, locked condition. A quick walk-around inspection at shutdown can catch new damage or leaks before the next shift.

#416 · Crane Operation

What is a swing lock and why is it engaged during shutdown?

Answer

A swing lock is a mechanical pin or brake that prevents the upper structure of the crane from rotating. It is engaged during shutdown and during travel so wind or an accidental control bump cannot cause the boom to swing unexpectedly. Leaving the swing unlocked overnight or in wind is a common cause of unattended crane incidents.

#417 · Crane Operation

Why should the boom be lowered rather than left raised when a crane is shut down for an extended period?

Answer

A raised boom presents more surface area to wind and creates a higher centre of gravity, both of which increase tipping risk during storms or high wind. Lowering the boom to a rest position or near-horizontal reduces this risk significantly. Manufacturer guidance usually specifies a maximum safe boom angle for unattended parking in wind.

#418 · Crane Operation

What minimum clearance must generally be maintained between a crane and energized overhead power lines?

Answer

Minimum clearance distances are set by regulation and increase with the voltage of the line, commonly starting around 3 metres for lower voltage lines and increasing for higher voltages. The exact distance must be confirmed for the specific voltage and jurisdiction before work begins. A dedicated spotter is often required whenever work occurs near power lines regardless of the calculated clearance.

#419 · Crane Operation

What should be done if the exact voltage of a nearby overhead power line is unknown?

Answer

The operator should treat the line as if it were carrying the highest voltage reasonably possible for that type of line and maintain the largest applicable clearance distance until the utility confirms the actual voltage. Work should not proceed based on a guess. Contacting the utility company directly is the correct way to confirm voltage and any required outage or clearance arrangements.

#420 · Crane Operation

What precaution should be taken when a crane is operating near other equipment or vehicles moving on the same site?

Answer

A clear exclusion zone around the crane's working radius should be established and communicated to all equipment operators and workers on site, often marked with barricades, cones, or signage. Ground crew should stay out of the swing radius and the path of the load. Radio coordination with other equipment operators helps prevent someone driving into the crane's working area unexpectedly.

#421 · Crane Operation

Why should ground workers never stand or walk under a suspended load?

Answer

Standing under a suspended load exposes a worker to being struck if the load shifts, the rigging fails, or the crane loses control of the load unexpectedly. This is one of the most consistently enforced rules in crane safety regulations across jurisdictions. Even brief loads and short lifts carry this same risk, so the rule applies at all times.

#422 · Crane Operation

What is the purpose of establishing a barricaded exclusion zone around a lift?

Answer

A barricaded exclusion zone keeps workers who are not directly involved in the lift out of the crane's swing radius and the load's travel path. It reduces the chance of someone wandering into a hazard area without realizing a lift is in progress. The zone should be sized to account for the full swing radius plus any load that could swing outward.

#423 · Crane Operation

What information does a load chart typically require the operator to know about the crane's setup before it can be used correctly?

Answer

The operator needs to know the exact boom length, boom angle or radius, counterweight configuration, and outrigger spread since load charts are specific to each combination. Using the wrong chart column for the actual configuration can give a false sense of safety margin. Checking the crane's actual outrigger and counterweight setup against the chart before lifting is a required step.

#424 · Crane Operation

What does 'working load limit' generally refer to when discussed alongside rigging and load charts?

Answer

Working load limit refers to the maximum load a specific piece of rigging equipment, such as a sling or shackle, is rated to carry safely under normal conditions. It differs from the crane's rated capacity from the load chart, which applies to the crane itself. Both the rigging's working load limit and the crane's chart capacity must be checked, and the lift is limited by whichever is lower.

#425 · Crane Operation

An operator notices the LMI display and the physical feel of the crane during a lift seem to disagree. What should the operator do?

Answer

The operator should treat this as a warning sign, stop the lift, and lower the load to a safe position rather than trusting either indicator alone. Electronic systems can fail or be miscalibrated, so unusual sounds, boom deflection, or outrigger movement should never be ignored even if the display looks normal. The crane should be inspected before further lifting resumes.

#426 · Crane Operation

What is the difference between a two-block condition and an overload condition?

Answer

A two-block condition happens when the hook block contacts the boom tip because the hoist was raised too far, and it can shear the wire rope. An overload condition happens when the load exceeds the crane's rated capacity for its current configuration, which can cause tipping or structural failure. Both are serious but are caused by different operator errors and are prevented by different safety devices.

#427 · Crane Operation

Why is it important for the operator to keep both hands and full attention on the controls during a live lift?

Answer

Coordinating hoist, swing, boom angle, and telescope functions smoothly requires continuous attention, and a distracted operator can miss an alarm or make an abrupt movement that swings the load. Crane controls respond with a delay due to hydraulic lag, so inputs must be anticipated and adjusted smoothly. Losing focus even briefly during a lift can lead to loss of load control.

#428 · Crane Operation

What should happen if the designated signal person needs to step away during an active lift?

Answer

The lift should be paused and the load set down or held stationary until a replacement signal person is clearly identified and confirmed with the operator. The operator should never continue taking signals from an unconfirmed or unofficial person. A brief pause to confirm the handoff prevents miscommunication mid-lift.

#429 · Crane Operation

What is meant by 'boom deflection' and why might an operator watch for it?

Answer

Boom deflection is the bending or flexing of the boom under load, which is normal to some degree but becomes a warning sign if it is excessive or sudden. Unusual deflection can indicate the crane is near or beyond its structural limits even if the LMI has not yet alarmed. Operators are trained to notice both the instrument readings and the physical behaviour of the machine.

#430 · Crane Operation

Why should outriggers always be fully extended and set on stable footing before a lift, even for a light load?

Answer

Load charts assume a specific outrigger configuration, usually fully extended, and using a shorter setting without adjusting to the correct reduced-capacity chart can lead to an unplanned overload situation. Unstable or unlevel footing can also cause the crane to settle unevenly during the lift, changing its effective stability. Proper cribbing or pads should be used on soft ground regardless of how light the load seems.

#431 · Crane Operation

What does it mean to operate a crane 'on rubber' versus 'on outriggers'?

Answer

Operating on rubber means lifting with the crane supported by its tires rather than outriggers, which is only allowed for specific crane models and load charts designed for that configuration. Operating on outriggers means the crane is supported by extended and set outrigger pads for greater stability and capacity. The correct load chart must match whichever configuration is actually being used.

#432 · Crane Operation

What hand signal typically indicates 'raise the boom'?

Answer

The raise boom signal is usually given with the arm extended and the thumb pointing upward, sometimes with a small circular motion, following the standardized signal chart for that jurisdiction. The exact form can vary slightly by region, so operators and signal persons must use the same recognized signal chart. A pre-lift signal review confirms both parties agree on the signals being used.

#433 · Crane Operation

Why is a common, pre-lift review of hand signals important even for experienced crews?

Answer

Different job sites or companies sometimes use slightly different signal charts, and a mismatch between operator and signal person expectations can cause a dangerous miscommunication. A short review before the lift confirms both parties are using the same signal set. This is especially important when working with an unfamiliar crew or a new signal person.

#434 · Crane Operation

What is a critical lift and how does its planning differ from a routine lift?

Answer

A critical lift is one that involves unusually high risk, such as lifting near or above 75 to 90 percent of rated capacity, a tandem lift, lifting over occupied areas, or lifting personnel. It requires a formal written lift plan, engineering review in some cases, and often a pre-lift meeting with all involved crew. Routine lifts still require planning but do not typically need this level of formal documentation.

#435 · Crane Operation

What should an operator verify about the ground beneath outrigger pads before setting up on a job site?

Answer

The operator should verify the ground can support the point loading of the outrigger pads without settling, checking for buried utilities, soft fill, slopes, or recent excavation nearby. Cribbing, timber mats, or larger pads should be used to spread the load over soft or uncertain ground. A site that looks solid on the surface can still have unstable fill underneath.

#436 · Crane Operation

What does 'percentage of rated capacity' displayed on an LMI screen actually represent?

Answer

It represents the current calculated load, including the hook block and rigging weight, as a percentage of the maximum capacity allowed for the crane's exact current configuration and radius. A reading of 100% means the crane is at its absolute limit for that setup, not that it still has extra margin. Operators are trained to stay well below 100% with a safety margin rather than working right up to the limit.

#437 · Crane Operation

Should the weight of rigging, such as slings and shackles, be included when calculating whether a lift is within capacity?

Answer

Yes, the total weight lifted by the crane includes the load itself plus the hook block, rigging, and any lifting attachments, all of which count against the rated capacity. Forgetting to include rigging weight can cause an operator to unknowingly exceed the chart capacity even if the load itself seemed within limits. Most LMI systems account for hook block weight automatically but rigging added below the hook must be factored in manually in some cases.

#438 · Crane Operation

What should an operator do if visibility drops sharply due to fog or heavy precipitation during an outdoor lift?

Answer

The operator should stop the lift and lower the load to the ground if visibility no longer allows clear observation of the load, signal person, and surrounding hazards. Continuing to operate with poor visibility increases the risk of striking an obstruction or losing track of ground crew. Operations should not resume until visibility returns to a safe level.

#439 · Crane Operation

What is the purpose of a pre-lift meeting for a planned crane operation?

Answer

A pre-lift meeting brings together the operator, signal person, riggers, and supervisor to review the lift plan, confirm the load weight, rigging method, signals, and any site hazards before work begins. It gives everyone a chance to raise concerns before the load is in the air rather than during the lift. This is especially important for critical or tandem lifts.

#440 · Crane Operation

What does 'centre of gravity' mean in relation to rigging a load correctly?

Answer

Centre of gravity is the point where a load's weight is balanced in all directions, and rigging must be attached so the crane hook lifts directly above this point. If the rigging point is offset from the centre of gravity, the load will tilt or swing unpredictably once it leaves the ground. Riggers often do a trial lift a short distance up to check balance before continuing.

#441 · Crane Operation

Why is a trial lift of a few centimetres off the ground recommended before a full lift begins?

Answer

A trial lift lets the crew check that the load is balanced, the rigging is secure, and the crane is stable before committing to a full lift. It gives a chance to stop and correct a problem while the load is still close to the ground and the risk of injury is lower. If anything looks wrong during the trial lift, the load should be set back down immediately.

#442 · Crane Operation

What should an operator do if a ground crew member enters the crane's swing radius without authorization during an active lift?

Answer

The operator should stop all motion immediately, even without a formal stop signal, since crane movement that could strike or crush a worker takes priority over completing the lift. The load should be held steady until the worker is confirmed clear of the hazard area. This falls under the same responsibility as obeying any emergency stop signal.

#443 · Crane Operation

Why do Red Seal standards require operators to understand both the load chart and the LMI system rather than relying on the LMI alone?

Answer

The LMI is an electronic aid that can fail, be miscalibrated, or display incorrect data if the crane configuration is entered wrong, so it should support the operator's own chart reading rather than replace it. An operator who understands the load chart can catch a discrepancy between what the LMI shows and what the actual setup allows. This layered approach is a core safety principle across Red Seal crane operator training.

#444 · Crane Operation

What does coordinated boom and swing movement mean during a lift?

Answer

It means moving the boom and slewing the upper together smoothly instead of one after the other. This keeps the load path predictable and reduces sudden swinging or pendulum motion. Jerky, uncoordinated inputs are one of the biggest causes of load swing accidents.

#445 · Crane Operation

Why should telescoping and hoisting never be combined carelessly on a live lift?

Answer

Extending the boom changes the radius, and radius changes the rated capacity from the load chart. If you telescope out while a load is suspended without checking the chart, you can exceed the crane's capacity at the new radius. Always confirm the new radius is still within rated capacity before or while telescoping under load.

#446 · Crane Operation

What is boom deflection?

Answer

Boom deflection is the natural bending or flexing of the boom under load, especially at longer lengths and lower angles. It is normal but must be accounted for because it changes the actual radius from what a static chart reading might suggest. Operators learn to anticipate it rather than fight it with sudden corrections.

#447 · Crane Operation

Scenario: an operator feels the joystick response lag slightly when starting a hoist. What should they do?

Answer

They should stop and report it rather than push through it. A lag in control response can point to a hydraulic or electronic control issue that could worsen mid-lift. Lifting operations should not continue until the cause is checked and cleared by a qualified person.

#448 · Crane Operation

Why do experienced operators favor slow, small inputs when starting a lift off the ground?

Answer

Small inputs let the operator feel how the load reacts before committing to full speed. This is when sling angle problems, an off-center load, or an unexpected snag are most likely to show up. Starting slow gives time to stop before a small problem becomes a swinging or dropped load.

#449 · Crane Operation

What is the difference between the structural chart and the capacity chart on a load chart page?

Answer

The capacity chart shows the maximum weight the crane can lift at a given radius and boom length based on tipping and rated limits. The structural chart shows the maximum weight the boom itself can handle before structural damage, which can be lower in some configurations. The operator must respect whichever number is lower.

#450 · Crane Operation

What is radius, as used on a load chart?

Answer

Radius is the horizontal distance from the crane's center of rotation to the center of the suspended load. As radius increases, rated capacity decreases even though the crane itself has not changed. It is one of the two main numbers, along with boom length or angle, used to read a load chart.

#451 · Crane Operation

Scenario: the crane is set up on outriggers fully extended, but the load chart lists a lower capacity for partial extension. Which number applies?

Answer

The number for the outrigger configuration actually in use applies, which in this case is full extension since that is how the crane is set up. Using the wrong outrigger column is a common and dangerous chart-reading error. The operator must confirm and match the actual rigging setup to the correct chart column every time.

#452 · Crane Operation

Why must the operator know the exact weight of the load before consulting the load chart?

Answer

The load chart only protects the crane if the input weight is accurate. If the actual load weight is unknown or underestimated, the operator could pick a radius that looks safe on paper but is not. Weight should come from a manifest, weight tag, or engineering calculation, never a guess.

#453 · Crane Operation

What must be added to the load weight when calculating total load for chart purposes?

Answer

The rigging weight (slings, shackles, spreader bars, headache ball or block) and any hook block weight must be added to the load weight. Ignoring rigging weight is a common way operators unknowingly exceed rated capacity. The chart capacity number already assumes some standard hook block weight, so extra rigging must be added on top.

#454 · Crane Operation

What is percentage of chart capacity, and why does it matter?

Answer

It is the actual load weight divided by the maximum rated capacity at that radius, expressed as a percentage. Most site safety rules cap routine lifts well below 100 percent, often at 75 to 85 percent, to leave a safety margin. Working close to 100 percent leaves no room for wind, dynamic loading, or small setup errors.

#455 · Crane Operation

What is the two-arm Emergency Stop signal, and how does it differ from a standard Stop?

Answer

The Emergency Stop signal uses both arms extended out to the sides, palms down, moved back and forth, distinguishing it from the one-arm standard Stop signal. It overrides any other instruction being given at the time and must be obeyed instantly by the operator. Unlike the routine Stop signal, anyone on site, not just the designated signal person, can give an Emergency Stop.

#456 · Crane Operation

Why should a crane operator only take signals from one designated signal person at a time?

Answer

Taking direction from multiple people creates confusion about whose signals are current and correct. Conflicting instructions can lead to the operator misjudging the load position or clearance. The one exception is the emergency stop signal, which anyone can give and the operator must always obey.

#457 · Crane Operation

Scenario: the operator loses visual contact with the signal person mid-lift. What should the operator do?

Answer

The operator should stop all motion immediately and hold until visual contact is restored or radio contact confirms it is safe to continue. Continuing to move a load without a clear line of communication to the signal person is not acceptable. Once contact is regained, the lift can resume with fresh signals.

#458 · Crane Operation

What is radio protocol for a blind lift where the operator cannot see the load?

Answer

The signal person gives continuous, clear verbal direction over radio, and the operator confirms each instruction back before acting. Standard phrasing and confirmed callbacks reduce the chance of a misheard direction. If radio communication is lost during a blind lift, the operator stops immediately.

#459 · Crane Operation

What qualifies someone to act as a signal person on a crane job site?

Answer

A signal person must be trained and, on many sites, certified in standard hand signals and radio communication, and must know the load chart basics well enough to recognize a dangerous situation. They also need a clear line of sight to both the load and the operator, or a reliable radio link. Untrained bystanders should never be relied on to give signals.

#460 · Crane Operation

Why is a pre-lift signal briefing important before a complex or blind lift?

Answer

It lets the operator and signal person agree in advance on which signals will be used, radio channel, and what to do if communication is lost. This avoids confusion once the lift is underway and everyone is busy watching the load. A short briefing takes a few minutes and prevents costly misunderstandings.

#461 · Crane Operation

What is an anti-two-block device?

Answer

It is a safety device that senses when the hook block or ball gets too close to the boom tip sheave and automatically stops the hoisting or telescoping motion that would cause a two-block condition. Two-blocking happens when the block is pulled up so far it jams into the boom tip, which can snap the hoist line or damage the boom. The device is a backup, not a replacement for operator attention.

#462 · Crane Operation

Scenario: the anti-two-block alarm sounds and the motion stops on its own. What should the operator do next?

Answer

The operator should reverse the motion that caused the alarm, moving the block away from the boom tip, rather than trying to force through the automatic stop. Forcing the system or bypassing it defeats the purpose of the device and risks a real two-block failure. Once clear, the operator should report the event so the device and rigging can be checked.

#463 · Crane Operation

Why must an anti-two-block device be tested before starting work each shift?

Answer

Like any safety device it can fail, get knocked loose, or have a damaged weight or cable, so a daily function test confirms it will actually work if needed. Skipping this check means the operator might not find out the device is inactive until an actual two-block event happens. Most manufacturers specify a simple pre-shift test procedure.

#464 · Crane Operation

Is it acceptable to disable or bypass an anti-two-block device to finish a job faster?

Answer

No. Bypassing this device is a serious violation and removes a critical protection against a sudden line failure or boom damage. If the device is malfunctioning, the crane should be taken out of service until it is repaired, not operated with the safety feature disabled. This applies even for a short, seemingly low-risk lift.

#465 · Crane Operation

What physically triggers a typical anti-two-block sensor?

Answer

A weighted trip line hangs near the boom tip sheave, and when the hook block rises far enough to lift and slacken that weight, it trips a switch that cuts hoist and telescope-out power. Some newer systems use proximity or cable-length sensors instead of a physical weight. Either way, the goal is the same: stop motion before contact happens.

#466 · Crane Operation

What is a Load Moment Indicator (LMI)?

Answer

An LMI is an electronic system that continuously calculates the crane's load moment (weight times radius) and compares it to rated capacity in real time, warning the operator as the crane approaches its limit. It typically uses sensors on boom angle, length, and load weight to do this calculation automatically. It gives audible and visual warnings before an overload condition is reached.

#467 · Crane Operation

What is a Rated Capacity Limiter (RCL) and how is it different from a simple indicator?

Answer

An RCL not only warns the operator like an LMI but also automatically cuts off motions that would increase the overload condition, such as hoisting up or extending the boom further out. It acts as an automatic backup if the operator does not stop in time. It does not replace the operator's responsibility to read the load chart and plan the lift correctly.

#468 · Crane Operation

Scenario: the LMI shows 95 percent of capacity and the alarm is chirping intermittently. What should the operator do?

Answer

The operator should treat this as a warning to stop increasing radius or load, not as permission to keep pushing toward the limit. Site rules and best practice usually call for staying at a lower percentage with margin for wind or dynamic effects. If the percentage keeps climbing, the operator should pause the lift and reassess before continuing.

#469 · Crane Operation

Why must LMI settings be reconfigured whenever the crane's rigging configuration changes?

Answer

The LMI calculates capacity based on programmed data for boom length, counterweight, outrigger spread, and other setup details. If the actual configuration does not match what is programmed into the LMI, its warnings and cutoffs will be based on wrong data and cannot be trusted. Operators must confirm and set the correct configuration in the LMI before every lift.

#470 · Crane Operation

Can an operator rely on the LMI alone instead of reading the printed load chart?

Answer

No. The LMI is a valuable backup and warning system, but the operator is still responsible for planning the lift using the printed load chart and knowing the numbers in advance. Electronic systems can have sensor errors, programming mistakes, or full failures. Treating the LMI as the only safeguard is a dangerous habit.

#471 · Crane Operation

What should an operator do first if engine power is suddenly lost while a load is suspended?

Answer

The operator should apply the hoist brake and swing brake to hold the load in place, keep hands off other controls, and avoid panic movements. Most cranes are designed so brakes hold the load automatically when power drops. The next step is to assess whether the load can be safely lowered using emergency or manual lowering procedures.

#472 · Crane Operation

What is the correct response to a hydraulic pressure loss warning while boom is extended and loaded?

Answer

The operator should stop all motion immediately and avoid any action that relies on hydraulic pressure to control a fall, such as free-falling the boom. Lowering the load slowly, if the system allows controlled lowering, is preferred over leaving a load hanging on a failing system. If safe lowering is not possible, the area below and around the load must be cleared and cordoned off while help is called.

#473 · Crane Operation

What does an overload alarm indicate and what is the immediate operator response?

Answer

It indicates the crane has reached or is approaching a load moment beyond safe rated capacity at the current radius and configuration. The operator should immediately stop any motion that increases the moment, such as hoisting up, booming out, or slewing to greater radius, and instead move toward reducing the moment. This usually means booming in or lowering the load rather than continuing the original motion.

#474 · Crane Operation

Scenario: a sudden gust of wind hits mid-lift and the load starts to swing hard. What is the priority?

Answer

The priority is to regain control of the load without making sudden corrective moves that add more swing energy. The operator should use small, opposing inputs to dampen the swing and may need to set the load down as soon as it is safe rather than fight a growing pendulum. If control cannot be regained safely, personnel below and around the load area must be warned immediately.

#475 · Crane Operation

Why should an operator never try to manually override a rated capacity limiter cutoff during an apparent false alarm?

Answer

What looks like a false alarm can sometimes be a real overload condition caused by a sensor reading the operator did not expect, such as a snagged load or wind loading. Overriding the system without confirming the actual load moment is safe removes the last automatic layer of protection. The correct step is to stop, investigate the cause, and only continue once the condition is confirmed safe.

#476 · Crane Operation

What is the general emergency procedure order when something goes wrong during a lift: stop, assess, or lower?

Answer

The general order is stop first, then assess the situation and hazards, then decide on lowering or another controlled action. Reacting immediately with a big motion before understanding the problem often makes things worse. This stop-assess-act sequence applies to almost every crane emergency, from power loss to overload alarms to sudden wind.

#477 · Crane Operation

What steps belong in a proper end-of-shift crane shutdown?

Answer

The boom should be lowered to a safe stowed position, the load block should be brought down and secured close to the ground or a designated rest point, and swing brake and parking brake should be applied. Outriggers are typically retracted only after the boom is stowed, following the manufacturer's sequence. The engine is shut off last, and keys or access controls are secured.

#478 · Crane Operation

Why should the hook block never be left hanging freely at height overnight?

Answer

A freely hanging block can swing in wind and strike the boom or nearby structures, and it presents an overhead hazard to anyone working below. It should be lowered and either rested on the ground, chained off, or otherwise secured against movement. This is a standard part of proper securing procedure at the end of any operating period.

#479 · Crane Operation

What is the purpose of applying the swing lock or swing brake during shutdown?

Answer

It prevents the upper structure from rotating on its own due to wind, ground slope, or vibration while the crane is unattended. An unlocked swing on a parked crane can allow the boom to drift into other equipment, structures, or overhead lines. Swing lock should be engaged as a routine part of every shutdown, even short breaks.

#480 · Crane Operation

Scenario: a crane is being shut down for the night on a site with expected high winds overnight. What extra step should the operator take?

Answer

The operator should check the manufacturer's out-of-service wind rating and, if winds are expected to exceed it, lower the boom further or place it in a specific storm position as specified in the operating manual. Some cranes require the boom laid down or telescoped in for high wind storage. Failing to prepare for forecasted high winds risks tip-over or structural damage even with the crane not operating.

#481 · Crane Operation

Why should outriggers stay extended and set until the boom is fully stowed, not retracted early?

Answer

The crane's stability depends on its outrigger footprint matching its configuration, and retracting outriggers while the boom is still up or extended can quickly create a tip-over condition. The correct shutdown sequence always stows the boom and load first, then retracts outriggers according to the manufacturer's steps. Skipping steps to save time is a common cause of shutdown accidents.

#482 · Crane Operation

What should be done with pendant lines, rigging, and loose gear before leaving a crane unattended?

Answer

All rigging should be removed from the hook or secured so nothing can swing, fall, or blow around, and loose tools or gear should be stowed in proper compartments or removed from the cab and deck. Leaving loose gear creates a falling object hazard and can also become a snag hazard the next time the crane operates. A clean, secured crane is part of a proper shutdown, not an optional extra.

#483 · Crane Operation

Why is a walk-around inspection part of proper shutdown, not just startup?

Answer

A shutdown walk-around lets the operator spot new damage, leaks, or loose fittings that happened during the day's work while it is still fresh and visible. Catching a problem at shutdown means it can be reported and addressed before the next shift starts, rather than being discovered mid-lift. It also confirms the crane was left in a safe, secured state.

#484 · Crane Operation

What is the minimum required clearance an operator must maintain from energized overhead power lines?

Answer

Minimum clearance depends on the voltage of the line and the applicable regulation, but it is never zero and increases with higher voltage. The operator must know the actual voltage and required clearance before starting any lift near power lines, not estimate by eye. A spotter and physical limiting devices are often required in addition to knowing the number.

#485 · Crane Operation

Scenario: another piece of mobile equipment is working close to the crane's swing radius. What should the operator do before starting to swing?

Answer

The operator should confirm with a spotter or radio that the swing path is clear and that the other equipment operator is aware the crane is about to move. Swinging without this confirmation risks a collision, since the other operator may not be watching the crane at all. On busy sites this kind of check should happen before every swing into a shared work zone, not just the first one.

#486 · Crane Operation

Why is a designated exclusion zone set up around a crane's working radius?

Answer

It keeps workers and equipment that are not part of the lift out of the area where a swinging load, boom failure, or tip-over could reach them. The zone is normally marked with barriers, tape, or signage and enforced by a signal person or spotter. Workers must never be allowed to walk or work under a suspended load, even briefly.

#487 · Crane Operation

What extra caution applies when operating a crane near an excavation or trench edge?

Answer

The added weight and vibration of the crane can cause the excavation wall to collapse or the ground under the outriggers to shift, so the crane should be set up at a safe distance based on the depth and soil type of the excavation. Outrigger mats or cribbing may need to be extended further from the edge than normal. A geotechnical or engineering assessment is often required before setting up close to a trench.

#488 · Crane Operation

Scenario: ground workers need to approach a load to guide it into place. When is it safe for them to do so?

Answer

It is only safe once the load is lowered close enough and stable enough that workers are not exposed to a swinging or falling load, and the signal person has confirmed it is safe to approach. Workers should never reach under or stand beneath a load that is still being lowered from height. Taglines are the preferred way to guide a load from a safe distance instead of hands-on contact while it is airborne.

#489 · Crane Operation

Why must an operator account for other equipment's blind spots, not just their own, when working in a shared area?

Answer

Other operators, such as those running excavators or trucks, may not be able to see the crane or its load from their cab position. Relying only on the crane operator's own visibility is not enough since a collision can happen from the other equipment moving into the crane's zone. Clear radio communication and site traffic control help cover blind spots on both sides.

#490 · Crane Operation

What is a tandem lift?

Answer

A tandem lift, also called a multi-crane lift, is when two or more cranes work together to lift and move a single load that is too large, long, or heavy for one crane alone. It requires careful load distribution planning between the cranes and tightly coordinated movement. It is considered a higher-risk lift category requiring extra planning and often engineering review.

#491 · Crane Operation

Why does a tandem lift require a written lift plan even when each crane's individual capacity is not exceeded?

Answer

Because the load share between cranes can shift unpredictably during the lift due to rigging geometry, ground conditions, or unequal crane movement, meaning one crane can end up carrying more than planned. A written plan calculates expected load share at each crane and sets limits and communication procedures in advance. Without this planning, an uneven load shift can overload one crane even if the total weight seemed within combined capacity.

#492 · Crane Operation

What is the role of a single lift director or designated coordinator in a multi-crane lift?

Answer

One person is designated to give direction to all cranes involved, rather than each operator responding to separate or conflicting signals. This person, usually with a full view of the whole operation, ensures both or all cranes move in a synchronized way. Multi-crane lifts without a single coordinating authority are far more prone to miscommunication and load imbalance.

#493 · Crane Operation

Scenario: during a tandem lift, one crane's operator notices their load is heavier than planned. What should happen?

Answer

The operator should stop immediately and communicate this to the lift coordinator rather than continuing or trying to compensate alone. Continuing without addressing the imbalance risks exceeding that crane's rated capacity even though the plan assumed a lower share. The lift should not resume until the load distribution issue is resolved or the plan is adjusted.

#494 · Crane Operation

Why must both cranes in a tandem lift move at matched, synchronized speeds?

Answer

If one crane hoists, booms, or swings faster than the other, the rigging angles change unevenly and can suddenly shift more load weight onto one crane or create dangerous side loading on the rigging. Synchronized, slow movement keeps the load level and keeps each crane's share close to what was planned. This is why tandem lifts are done at much slower speeds than a typical single-crane lift.

#495 · Crane Operation

What ground and setup consideration is especially critical for multi-crane lifts?

Answer

Both cranes must be on stable, level, properly matted ground since uneven settling between the two cranes during the lift can shift load distribution or change rigging angles unexpectedly. Ground conditions should be checked and prepared to the same standard for each crane involved, not just the one carrying the larger apparent share. A failure at one crane's setup can compromise the whole tandem lift.

#496 · Crane Operation

What is the safe response if wind speed approaches the manufacturer's rated operating limit during a lift in progress?

Answer

The operator should complete the current motion in a controlled way and set the load down as soon as safely possible rather than starting new lifts or continuing indefinitely. Continuing to operate above the rated wind limit risks tip-over or loss of load control. Once the limit is reached or expected to be exceeded, all crane operations should stop until conditions improve.

#497 · Crane Operation

Why is lightning in the area a reason to stop crane operations entirely, not just pause?

Answer

The crane's boom and cable act as a tall conductive path, making it a lightning strike risk to the operator and anyone nearby. Unlike wind, which can be monitored and worked around gradually, lightning requires a full stop and moving personnel to a safe location until the storm passes. Operations should not resume until the storm has cleared based on site lightning safety procedures.

#498 · Crane Operation

How does cold weather affect crane hydraulic and mechanical systems, and what should the operator do about it?

Answer

Cold temperatures thicken hydraulic fluid and can make components respond more slowly or stiffly than normal, and metal parts become more brittle. Operators should follow manufacturer warm-up procedures before working at full speed or capacity in cold weather. Skipping warm-up in extreme cold increases the risk of a sudden component failure.

#499 · Crane Operation

What visibility problem does fog or heavy precipitation create for crane operations, and how should it be handled?

Answer

Reduced visibility can prevent the operator from seeing the signal person, the load, or hazards in the swing path clearly enough to operate safely. If visibility drops below the point where the operator can reliably see required signals or the load path, operations should be paused. Radio communication alone is not always considered sufficient replacement for lost visual contact in these conditions.

#500 · Crane Operation

Scenario: a rapid temperature swing causes visible ice to form on the boom and rigging overnight. What must happen before lifting resumes?

Answer

The crane and rigging must be inspected and any ice cleared or confirmed not to affect operation before any lift begins, since ice adds unplanned weight and can make surfaces slippery for anyone working at height. Icy rigging can also affect how a sling grips or seats properly. Operating with ice still present risks both a miscalculated load weight and a fall hazard for ground crew.